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Location determination system and location determination method

Abstract

An E2EO module determines, in response to determination that a new element linked to an already installed element included in the communication system is to be added to the communication system, a location at which the new element is to be constructed based on new topology data including the new element and a current location of the already installed element indicated by location data.

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U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2018/0324219	12/2017	Xie	N/A	H04W 12/06
2020/0013107	12/2019	Kusano	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2018/181826	12/2017	WO	N/A

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Background/Summary

RELATED APPLICATIONS

(1) The present application is a National Phase of International Application Number PCT/JP2021/036885, filed Oct. 5, 2021.

TECHNICAL FIELD

(2) The present invention relates to a location determination system and a location determination method.

BACKGROUND ART

(3) As an example of a technology relating to management of virtualized network functions (VNFs) and other elements included in a communication system, in Patent Literature 1, it is described that a network function virtualization orchestrator (NFVO) performs healing and scaling on VNFs.

(4) In addition, there has been known a technology for determining a location of an element to be newly constructed by, for example, scale-out, at random or in a round robin.

CITATION LIST

Patent Literature

(5) [Patent Literature 1] WO 2018/181826 A1

SUMMARY OF INVENTION

Technical Problem

(6) With the technology as described in Patent Literature 1, when a location of an element to be newly constructed is determined at random or in a round robin, there is a fear in that the element may be constructed at an undesired location. For example, there is a fear in that an access and mobility management function (AMF) to be newly added under the control of a certain core

network system may be constructed at a location far from an already installed session management function (SMF) under the control of the core network system. Then, as a result, there is a fear in that there may occur performance degradation including deterioration of latency and reduction of throughput.

(7) The present invention has been made in view of the above-mentioned circumstances, and has an object to provide a location determination system and a location determination method which are capable of appropriately determining a location at which a new element is to be constructed in a communication system.

Solution to Problem

(8) In order to solve the above-mentioned problem, according to one embodiment of the present invention, there is provided a location determination system including: topology data storage means for storing topology data indicating a current status of a link between elements included in a communication system; location data storage means for storing location data indicating a current location of an element included in the communication system; and location determination means for determining, in response to determination that a new element linked to an already installed element included in the communication system is to be added to the communication system, a location at which the new element is to be constructed based on new topology data including the new element and the current location of the already installed element indicated by the location data.

(9) In one aspect of the present invention, the location determination means is configured to determine a location of a cluster in which the new element is to be constructed.

(10) In this aspect, the location determination means may be configured to determine a location different from a location of a cluster in which the already installed element linked to the new element in the topology data is constructed as the location of the cluster in which the new element is to be constructed.

(11) Further, in one aspect of the present invention, the location determination means is configured to determine a location of a server on which the new element is to be constructed.

(12) In this aspect, the location determination means may be configured to determine a location different from a location of a server on which the already installed element linked to the new element in the topology data is constructed as the location of the server in which the new element is to be constructed.

(13) Further, in one aspect of the present invention, the location determination means is configured to determine a location of a data center in which the new element is to be constructed.

(14) In this aspect, the location determination means may be configured to determine a location different from a location of a data center in which the already installed element linked to the new element in the topology data is constructed as the location of the data center in which the new element is to be constructed.

(15) Further, in one aspect of the present invention, functional units under control of a common network service are linked to each other in the topology data, the location data indicates a current location of an already installed functional unit under control of an already installed network service, and the location determination means is configured to determine, in response to determination that a new functional unit is to be added under the control of the already installed network service included in the communication system, a location at which the new functional unit is to be constructed based on the current location of the already installed functional unit indicated as the functional unit under the control of the already installed network service by the topology data, the current location being indicated by the location data.

(16) Further, in one aspect of the present invention, functional units belonging to a common network slice are linked to each other in the topology data, the location data indicates a current location of an already installed functional unit belonging to an already installed network slice, and the location determination means is configured to determine, in response to determination that a new functional unit belonging to the already installed network slice included in the communication

system is to be added to the communication system, a location at which the new functional unit is to be constructed based on the current location of the already installed functional unit indicated as the functional unit belonging to the already installed network slice by the topology data, the current location being indicated by the location data.

(17) Further, in one aspect of the present invention, the location determination means is configured to determine a location satisfying a given performance requirement relating to communication between the already installed element and the new element as the location at which the new element is to be constructed.

(18) Further, in one aspect of the present invention, the location determination system further includes constructing means for constructing the new element at the determined location.

(19) Further, according to one embodiment of the present invention, there is provided a location determination method including the steps of: storing, in topology data storage means, topology data indicating a current status of a link between elements included in a communication system; storing, in location data storage means, location data indicating a current location of an element included in the communication system; and determining, in response to determination that a new element linked to an already installed element included in the communication system is to be added to the communication system, a location at which the new element is to be constructed based on new topology data including the new element and the current location of the already installed element indicated by the location data.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram for illustrating an example of a communication system in one embodiment of the present invention.

(2) FIG. 2 is a diagram for illustrating an example of the communication system in the one embodiment of the present invention.

(3) FIG. 3 is a diagram for schematically illustrating an example of a functional unit group constructed in the communication system in the one embodiment of the present invention.

(4) FIG. 4 is a diagram for schematically illustrating an example of a network slice to which functional units illustrated in FIG. 3 belong.

(5) FIG. 5 is a diagram for illustrating an example of links between elements constructed in the communication system in the one embodiment of the present invention.

(6) FIG. 6 is a diagram for illustrating an example of data structure of a bundle file.

(7) FIG. 7 is a diagram for illustrating an example of an onboarding screen.

(8) FIG. 8 is a functional block diagram for illustrating an example of functions implemented by an MPS and an NOS in the one embodiment of the present invention.

(9) FIG. 9 is a diagram for illustrating an example of data structure of a data group generated based on the bundle file.

(10) FIG. 10 is a diagram for illustrating an example of data structure of physical inventory data.

(11) FIG. 11 is a diagram for schematically illustrating an example of topology data.

(12) FIG. 12 is a diagram for schematically illustrating an example of the topology data.

(13) FIG. 13 is a diagram for schematically illustrating an example of the topology data.

(14) FIG. 14 is a diagram for schematically illustrating an example of the topology data.

(15) FIG. 15 is a diagram for illustrating an example of a configuration of an E2EO module.

(16) FIG. 16 is a diagram for schematically illustrating an example of an event occurrence situation.

(17) FIG. 17 is a diagram for schematically illustrating an example of the event occurrence situation.

- (18) FIG. **18** is a diagram for schematically illustrating an example of a network service group and a functional unit group that are constructed in a core network system in the one embodiment of the present invention.
- (19) FIG. **19** is a diagram for schematically illustrating an example of the topology data.
- (20) FIG. **20** is a diagram for schematically illustrating an example of the topology data.
- (21) FIG. **21A** is a flow chart for illustrating an example of a flow of processes performed by a vendor terminal, the MPS, and the NOS in the one embodiment of the present invention.
- (22) FIG. **21B** is a flow chart for illustrating an example of the flow of the processes performed by the vendor terminal, the MPS, and the NOS in the one embodiment of the present invention.
- (23) FIG. **22** is a flow chart for illustrating an example of a flow of processes performed by a purchaser terminal, the MPS, and the NOS in the one embodiment of the present invention.
- (24) FIG. **23A** is a flow chart for illustrating an example of a flow of processes performed by the NOS in the one embodiment of the present invention.
- (25) FIG. **23B** is a flow chart for illustrating an example of the flow of the processes performed by the NOS in the one embodiment of the present invention.
- (26) FIG. **23C** is a flow chart for illustrating an example of the flow of the processes performed by the NOS in the one embodiment of the present invention.
- (27) FIG. **23D** is a flow chart for illustrating an example of the flow of the processes performed by the NOS in the one embodiment of the present invention.
- (28) FIG. **23E** is a flow chart for illustrating an example of the flow of the processes performed by the NOS in the one embodiment of the present invention.
- (29) FIG. **23F** is a flow chart for illustrating an example of the flow of the processes performed by the NOS in the one embodiment of the present invention.
- (30) FIG. **24** is a flow chart for illustrating an example of a flow of a process performed by the NOS in the one embodiment of the present invention.
- (31) FIG. **25** is a flow chart for illustrating an example of a flow of a process performed by the NOS in the one embodiment of the present invention.
- (32) FIG. **26** is a flow chart for illustrating an example of a flow of a process performed by the NOS in the one embodiment of the present invention.
- (33) FIG. **27** is a flow chart for illustrating an example of a flow of a process performed by the NOS in the one embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

(34) [Overall Configuration]

- (35) One embodiment of the present invention is now described in detail with reference to the drawings.
- (36) FIG. **1** and FIG. **2** are each a diagram for illustrating an example of a communication system **1** in the one embodiment of the present invention. FIG. **1** is an illustration drawn with attention being given to locations of a data center group included in the communication system **1**. FIG. **2** is an illustration drawn with attention being given to various computer systems implemented in the data center group included in the communication system **1**.
- (37) As illustrated in FIG. **1**, the data center group included in the communication system **1** is classified into central data centers **10**, regional data centers **12**, and edge data centers **14**.
- (38) For example, several central data centers **10** are dispersedly arranged in an area (for example, in Japan) covered by the communication system **1**.
- (39) For example, tens of regional data centers **12** are dispersedly arranged in the area covered by the communication system **1**. For example, when the area covered by the communication system **1** is the entire area of Japan, one or two regional data centers **12** may be arranged in each prefecture.
- (40) For example, thousands of edge data centers **14** are dispersedly arranged in the area covered by the communication system **1**. In addition, each of the edge data centers **14** can communicate to/from a communication facility **18** provided with an antenna **16**. In this case, as illustrated in FIG.

1, one edge data center **14** may be capable of communicating to/from several communication facilities **18**. The communication facility **18** may include a computer, for example, a server computer. The communication facility **18** in this embodiment performs radio communication to/from a user equipment (UE) **20** via the antenna **16**.

(41) A plurality of servers are arranged in each of the central data centers **10**, the regional data centers **12**, and the edge data centers **14** in this embodiment.

(42) In this embodiment, for example, the central data centers **10**, the regional data centers **12**, and the edge data centers **14** can communicate to/from one another. Communication can also be performed between the central data centers **10**, between the regional data centers **12**, and between the edge data centers **14**.

(43) As illustrated in FIG. 2, the communication system **1** in this embodiment includes a marketplace system (MPS) **30**, a network operation system (NOS) **32**, a plurality of purchaser terminals **34**, a plurality of vendor terminals **36**, a plurality of radio access networks (RANs) **38**, a plurality of core network systems **40**, and a plurality of UEs **20**. The purchaser terminal **34** and the vendor terminal **36** are also illustrated in FIG. 1. The core network system **40**, the RAN **38**, and the UE **20** cooperate with each other to implement a mobile communication network.

(44) The RAN **38** is a computer system, which is provided with the antenna **16**, and corresponds to an eNodeB (eNB) in 4G and an NR base station (gNB) in 5G. The RANs **38** in this embodiment are implemented mainly by server groups arranged in the edge data centers **14** and the communication facilities **18**. A part of the RAN **38** (for example, virtual distributed unit (vDU) or virtual central unit (vCU) in 4G or distributed unit (DU) or central unit (CU) in 5G) may be implemented by the central data center **10** or the regional data center **12** instead of the edge data center **14**.

(45) The core network system **40** is a system corresponding to an evolved packet core (EPC) in a fourth generation mobile communication system (hereinafter referred to as “4G”) or a 5G core (5GC) in a fifth generation mobile communication system (hereinafter referred to as “5G”). The core network systems **40** in this embodiment are implemented mainly by server groups arranged in the central data centers **10** or the regional data centers **12**.

(46) The MPS **30** in this embodiment is configured, for example, on a cloud platform and includes a processor **30a**, a storage unit **30b**, and a communication unit **30c**, as illustrated in FIG. 2. The processor **30a** is a program control device, for example, a microprocessor which operates in accordance with a program installed in the MPS **30**. The storage unit **30b** is, for example, a storage element such as a ROM or RAM, a solid state drive (SSD), a hard disk drive (HDD), or the like. The storage unit **30b** stores a program to be executed by the processor **30a**, and the like. The communication unit **30c** is, for example, a communication interface such as a network interface card (NIC) or a wireless LAN module. Software-defined networking (SDN) may be implemented in the communication unit **30c**. The communication unit **30c** exchanges data with the NOS **32** and the purchaser terminal **34**.

(47) It is assumed that the MPS **30** in this embodiment is implemented by a server group arranged in the central data center **10**. The MPS **30** may be implemented by a server group arranged in the regional data center **12**.

(48) The NOS **32** in this embodiment is configured, for example, on a cloud platform and includes a processor **32a**, a storage unit **32b**, and a communication unit **32c**, as illustrated in FIG. 2. The processor **32a** is a program control device, for example, a microprocessor which operates in accordance with a program installed in the NOS **32**. The storage unit **32b** is, for example, a storage element such as a ROM or RAM, a solid state drive (SSD), a hard disk drive (HDD), or the like. The storage unit **32b** stores a program to be executed by the processor **32a**, and the like. The communication unit **32c** is, for example, a communication interface such as a NIC or a wireless LAN module. Software-defined networking (SDN) may be implemented in the communication unit **32c**. The communication unit **32c** exchanges data with the MPS **30**, the vendor terminal **36**, the

RAN **38**, and the core network system **40**.

(49) In this embodiment, the NOS **32** is implemented by a server group arranged in the central data center **10**. The NOS **32** may be implemented by a server group arranged in the regional data center **12**.

(50) In this embodiment, for example, in response to a purchase request for a network service (NS) by a purchaser, the network service for which the purchase request has been made is constructed in the RAN **38** or the core network system **40**. Then, the constructed network service is provided to the purchaser.

(51) For example, a network service such as a voice communication service, a data communication service, or the like is provided to the purchaser who is a mobile virtual network operator (MVNO). The voice communication service or the data communication service provided in this embodiment is eventually provided to a customer (end user) for the purchaser (MVNO in the above-mentioned example), who uses the UE **20** illustrated in FIG. **1** and FIG. **2**. The end user can perform voice communication or data communication to/from other users via the RAN **38** or the core network system **40**. The UE **20** of the end user can also access a data network, for example, the Internet, via the RAN **38** or the core network system **40**.

(52) In addition, in this embodiment, an IoT service may be provided to an end user who uses a robot arm, a connected car, or the like. In this case, an end user who uses, for example, a robot arm, a connected car, or the like may be a purchaser of the network service in this embodiment.

(53) In this embodiment, a container-type application execution environment, for example, Docker, is installed in the servers arranged in the central data center **10**, the regional data center **12**, and the edge data center **14**, and containers can be deployed in those servers and operated.

(54) The network service provided to the purchaser in this embodiment is formed of one or a plurality of functional units (for example, network function (NF)). In this embodiment, the functional unit is implemented by a containerized network function (CNF) being a container-based functional unit.

(55) The purchaser terminal **34** in this embodiment is, for example, a general computer used by the purchaser described above, such as a smartphone, a tablet terminal, or a personal computer.

(56) In this embodiment, for example, the above-mentioned purchaser can purchase a network service through a screen displayed on the purchaser terminal **34**. In this case, the purchaser can input service requirements for a network service to be purchased on the screen. For example, the purchaser can input, on the screen, service requirements including the type of service (for example, voice communication service, data communication service, or IoT service), a service scale (for example, the number of subscribers), whether or not to allow shared slices, an opposite IP, a target area, and the like. The purchaser can also input a password on the screen.

(57) In this embodiment, for example, when a predetermined purchase operation is executed after service requirements and a password are input, service requirement data indicating the input service requirements and password is transmitted from the purchaser terminal **34** to the MPS **30**.

(58) Then, the MPS **30** cooperates with the NOS **32** to construct, based on the service requirement data, a network service that satisfies the service requirements indicated by the service requirement data, and the constructed network service is provided to the purchaser.

(59) As described above, according to this embodiment, a network service that satisfies various purchaser's needs is flexibly constructed. Without being aware of the detailed implementation of the network service, the purchaser can receive the provision of a desired network service by only specifying some service requirements.

(60) FIG. **3** is a diagram for schematically illustrating an example of a functional unit group constructed in the communication system **1** in this embodiment. FIG. **4** is a diagram for schematically illustrating an example of a network slice to which functional units illustrated in FIG. **3** belong. The functional unit groups in 5G are illustrated in FIG. **3** and FIG. **4**, but functional unit groups in 4G may be constructed in the communication system **1** in this embodiment.

(61) In the example of FIG. 3, a gNB 56 including a DU 50, a central unit-control plane (CU-CP) 52, a central unit-user plane (CU-UP) 54a, a CU-UP 54b, and a CU-UP 54c is illustrated. In this embodiment, for example, it is assumed that a plurality of gNBs including this gNB 56 are arranged in a certain edge data center 14.

(62) Further, as illustrated in FIG. 3, an access-and-mobility management function (AMF) 60, a session management function (SMF) 62, a user plane function (UPF) 64a, and a UPF 64b are arranged in the central data center 10, and a UPF 64c is arranged in the regional data center 12.

(63) Further, in this embodiment, the UE 20 can also access a service 66a implemented in the central data center 10 via the UPF 64a. The UE 20 can also access a service 66b implemented in the central data center 10 via the UPF 64b. The UE 20 can also access a service 66c implemented in the regional data center 12 via the UPF 64c.

(64) As illustrated in FIG. 4, the DU 50, the CU-UP 54a, and the UPF 64a belong to a network slice 68a. Meanwhile, the DU 50, the CU-UP 54b, and the UPF 64b belong to a network slice 68b and a network slice 68c. The DU 50, the CU-UP 54c, and the UPF 64c belong to a network slice 68d. In FIG. 4, it is schematically illustrated that one DU 50, one CU-UP 54, and one UPF 64 belong to each network slice 68, but in general, a plurality of DUs 50, a plurality of CU-UPs 54, and a plurality of UPFs 64 belong to one network slice 68.

(65) FIG. 5 is a diagram for schematically illustrating an example of links between elements constructed in the communication system 1 in this embodiment.

(66) As illustrated in FIG. 5, a network service (NS), a network function (NF), a containerized network function component (CNFC), a pod, and a container have a hierarchical structure.

(67) The NS corresponds to an element having a granularity, for example, a 5GC, an EPC, a 5G RAN (gNB), or a 4G RAN (eNB).

(68) In 5G, the NF corresponds to an element having a granularity, for example, the AMF 60, the SMF 62, the UPF 64, the CU-CP 52, the CU-UP 54, or the DU 50. In 4G, the NF corresponds to an element having a granularity, for example, a mobility management entity (MME), a home subscriber server (HSS), a serving gateway (S-GW), a vDU, or a vCU. In this embodiment, for example, one NS includes one or a plurality of NFs. That is, one or a plurality of NFs are under the control of one NS.

(69) The CNFC corresponds to an element having a granularity, for example, DU mgmt or DU processing. The CNFC may be a microservice deployed on a server as one or more containers. For example, some CNFCs may be microservices that provide a part of the functions of the DU 50, the CU-CP 52, the CU-UP 54, and the like. Some CNFCs may be microservices that provide a part of the functions of the AMF 60, the SMF 62, the UPF 64, and the like. In this embodiment, for example, one NF includes one or a plurality of CNFCs. That is, one or a plurality of CNFCs are under the control of one NF.

(70) The pod refers to, for example, the minimum unit for managing a Docker container by Kubernetes. In this embodiment, for example, one CNFC includes one or a plurality of pods. That is, one or a plurality of pods are under the control of one CNFC.

(71) In this embodiment, for example, one pod includes one or a plurality of containers. That is, one or a plurality of containers are under the control of one pod.

(72) In addition, as illustrated in FIG. 5, a network slice (NSI) and a network slice subnet instance (NSSI) have a hierarchical structure.

(73) The NSIs can be said to be end-to-end virtual circuits that span a plurality of domains (for example, from the RAN 38 to the core network system 40). Each NSI may be a slice for high-speed and high-capacity communication (for example, eMBB), a slice for high-reliability and low-latency communication (for example, URLLC), or a slice for connecting a large quantity of terminals (for example, mMTC). The NSSIs can be said to be single domain virtual circuits dividing an NSI. Each NSSI may be a slice of a RAN domain, a slice of a mobile back haul (MBH) domain, or a slice of a core network domain.

(74) In this embodiment, for example, one NSI includes one or a plurality of NSSIs. That is, one or a plurality of NSSIs are under the control of one NSI. In this embodiment, a plurality of NSIs may share the same NSSI.

(75) In addition, as illustrated in FIG. 5, the NSSI and the NS generally have a many-to-many relationship.

(76) In addition, as described with reference to FIG. 4, in this embodiment, for example, one NF can belong to one or a plurality of network slices. Specifically, for example, network slice selection assistance information (NSSAI) including one or a plurality of pieces of sub-network slice selection assist information (S-NSSAI) can be set for one NF. In this case, the S-NSSAI is information associated with the network slice. The NF is not required to belong to the network slice.

(77) The vendor terminal **36** in this embodiment is a general computer used by a vendor, for example, a service provider related to the network service, such as a smartphone, a tablet terminal, a personal computer, or the like.

(78) In this embodiment, the vendor is provided with a continuous integration (CI)/continuous delivery (CD) pipeline including a development environment, a verification environment, and a test environment. In this embodiment, a verified bundle file corresponding to the network service to be provided to the purchaser, which is created by the vendor, is on-boarded by an onboarding process utilizing the CI/CD pipeline.

(79) The bundle file in this embodiment is, for example, a file obtained by compressing a file group having a predetermined directory structure (for example, a file having tar.gz format).

(80) FIG. 6 is a diagram for illustrating an example of data structure of the bundle file in this embodiment. As illustrated in FIG. 6, the bundle file in this embodiment includes business section data, technology section data, security section data, and operation section data.

(81) The business section data indicates business requirements of the network service such as the name of the network service, license requirements, and the definition of service level agreement (SLA). Further, the business section data in this embodiment includes data indicating mandatory input items and optional input items for the service requirements of the network service.

(82) The technology section data indicates, for example, the configuration of each element described with reference to FIG. 5. The technology section data also includes template data indicating a workflow script, various logics, and the like which are described below.

(83) The security section data indicates, for example, the security definition of the network service, for example, installation credentials.

(84) The operation section data indicates, for example, monitoring policies relating to network services, for example, a performance index value (for example, key performance indicator (KPI) or metric) of a monitoring target and a monitoring interval.

(85) The performance index values indicated by the operation section data may include performance index values regarding performance indices described in, for example, “TS 28.552, Management and orchestration; 5G performance measurements” or “TS 28.554, Management and orchestration; 5G end to end Key Performance Indicators (KPI).”

(86) The operation section data may also include, for example, monitoring script data indicating a monitoring script. The monitoring script data may include calculation logic data indicating a calculation logic for calculating a desired performance index value.

(87) The operation section data may also include a script template (Helm chart), a container image, and template data which are described below.

(88) FIG. 7 is a diagram for illustrating an example of an onboarding screen displayed on the vendor terminal **36** in this embodiment. In this embodiment, when the vendor specifies a path in which the bundle file is arranged and then clicks an “ONBOARDING” button **70**, the bundle file becomes on-boarded.

(89) As described above, in this embodiment, the vendor can easily perform onboarding of the

network service without being aware of the actual location at which a developed file group is onboarded.

(90) The functions of the MPS **30** and the NOS **32** in this embodiment and processes to be executed by the MPS **30** and the NOS **32** are further described in the following.

(91) FIG. **8** is a functional block diagram for illustrating an example of functions implemented by the MPS **30** and the NOS **32** in this embodiment. The plurality of functional blocks illustrated in the block diagram of the present application can be configured from, in terms of hardware, a circuit block, a memory, and other LSIs, and in terms of software, can be implemented by the CPU executing a program loaded onto the memory. Consequently, a person skilled in the art would understand that those functional blocks can be implemented in various forms by only hardware, by only software, or by a combination of hardware and software. The MPS **30** and the NOS **32** in this embodiment are not required to implement all of the functions illustrated in FIG. **8** and may implement functions other than those illustrated in FIG. **8**.

(92) As illustrated in FIG. **8**, the MPS **30** functionally includes, for example, a product management module **80**, a product catalog storage **82**, and a purchase management module **84**.

(93) The product management module **80** and the purchase management module **84** are implemented mainly by the processor **30a** and the communication unit **30c**. The product catalog storage **82** is implemented mainly by the storage unit **30b**.

(94) The above-mentioned functions may be implemented by executing, by the processor **30a**, a program that is installed in the MPS **30**, which is a computer, and that includes instructions corresponding to the above-mentioned functions. This program may be supplied to the MPS **30** via a computer-readable information storage medium such as an optical disc, a magnetic disk, a magnetic tape, a magneto-optical disc, a flash memory, or the like, or via the Internet or the like.

(95) In addition, as illustrated in FIG. **8**, for example, the NOS **32** functionally includes a bundle development module **90**, a service management module **92**, a service catalog storage **94**, an orchestration (end-to-end-orchestration (E2EO)) module **96**, an inventory management module **98**, a configuration management module **100**, a monitoring management module **102**, a security setting module **104**, a plurality of container management modules **106**, a repository module **108**, an inventory database **110**, and a bare metal management module **112**. In addition, the E2EO module **96** includes an active inventory **114**.

(96) The bundle development module **90** is implemented mainly by the processor **32a** and the communication unit **32c**. The service management module **92** and the bare metal management module **112** are each implemented mainly by the processor **32a**. The service catalog storage **94**, the repository module **108**, and the inventory database **110** are each implemented mainly by the storage unit **32b**. The E2EO module **96** is implemented mainly by the processor **32a**, the storage unit **32b**, and the communication unit **32c**. The inventory management module **98**, the configuration management module **100**, the monitoring management module **102**, the security setting module **104**, and the container management module **106** are each implemented mainly by the processor **32a** and the storage unit **32b**.

(97) The above-mentioned functions may be implemented by executing, by the processor **32a**, a program that is installed in the NOS **32**, which is a computer, and that includes instructions corresponding to the above-mentioned functions. This program may be supplied to the NOS **32** via a computer-readable information storage medium such as an optical disc, a magnetic disk, a magnetic tape, a magneto-optical disc, a flash memory, or the like, or via the Internet or the like.

(98) Further, FIG. **8** also shows a plurality of servers **120** included in the RANs **38** and the core network systems **40** illustrated in FIG. **2** and dispersedly arranged at various locations. Each of the plurality of container management modules **106** in this embodiment is associated with a server group that is a part of the plurality of servers **120**.

(99) In each of the plurality of container management modules **106** in this embodiment, for example, a container management tool such as Kubernetes, and a package manager such as Helm

are installed. The container management module **106** executes life cycle management of a container including the construction of the container such as the deployment and setting of the container for a server group (a plurality of servers **120**) associated with the container management module **106**.

(100) The container management module **106** is not required to be included in the NOS **32**. The container management module **106** may be provided in, for example, a server **120** (that is, the RAN **38** or the core network system **40**) managed by the container management module **106**, or a server that is annexed to the server **120**.

(101) For example, in this embodiment, the bundle development module **90** receives a bundle file from the vendor terminal **36**. Then, for example, in this embodiment, the bundle development module **90** generates a data group having data structure illustrated in FIG. **9**, based on the received bundle file. The data group illustrated in FIG. **9** is obtained by reconstructing the contents of the bundle file received by the bundle development module **90**.

(102) As illustrated in FIG. **9**, the data group generated by the bundle development module **90** includes product catalog data, service catalog data, inventory template data, CM template data, monitoring script data, security script data, Helm chart data, and container image data.

(103) The product catalog data is, for example, data corresponding to business section data included in a bundle file. As described above, the product catalog data indicates information regarding business requirements of the network service, such as the name of the network service displayed on the screen of the purchaser terminal **34**, license requirements, and the definition of the service level agreement (SLA).

(104) Further, the product catalog data in this embodiment includes data indicating mandatory input items and optional input items for the service requirements of the network service. In this embodiment, a screen to be displayed on the purchaser terminal **34** is generated based on, for example, the product catalog data.

(105) The service catalog data is, for example, data corresponding to a part of the technology section data included in the bundle file. The service catalog data contains a workflow script for constructing the network service.

(106) The service catalog data may also include information relating to a threshold value (for example, threshold value for abnormality detection) to be compared to the calculated performance index value.

(107) Further, the service catalog data may include requirement configuration correspondence data indicating the correspondence between a value of the above-mentioned service requirement data and the configuration of a functional unit group (for example, NF group) constructed in response to a purchase request.

(108) For example, the service catalog data may include requirement configuration correspondence data indicating the correspondence between the value of the service requirement data, the type of a functional unit group, and the number of functional units for each type. The requirement configuration correspondence data may indicate correspondences between, for example, “the number n1 of subscribers and one UPF,” “the number n2 of subscribers and one SMF,” “the number n3 of subscribers and one AMF,” “the number n4 of subscribers and one CU-CP,” “the number n5 of subscribers and one CU-UP,” and “the number n6 of subscribers and one DU.” What is associated with the service requirement data is not limited to the type and the number of 5G components, and the service requirement data and the type and the number of 4G components may be associated with each other.

(109) Further, for example, the requirement configuration correspondence data may indicate the correspondence between the value of the service requirement data and a location at which each functional unit included in a functional unit group constructed in response to a purchase request is to be constructed. In this case, the location associated with the value of the service requirement data in the requirement configuration correspondence data may be different depending on functional

units included in the functional unit group to be constructed.

(110) In addition, the service catalog data may include, for example, service template data indicating the logic to be used by the E2EO module **96**, which includes information required for the E2EO module **96** to construct the network service. Specifically, for example, the service template data includes information defining the NS, the NF, and the CNFC, and information indicating an NS-NF-CNFC correspondence relationship.

(111) The service catalog data may also include, for example, a slice template indicating the logic to be used by the E2EO module **96**, which includes information required for the E2EO module **96** to execute the instantiation of the network slice.

(112) The slice template data includes information on a “generic network slice template” defined by the GSM Association (GSMA) (“GSM is a trademark”). Specifically, the slice template data includes network slice template data (NST), network slice subnet template data (NSST), and network service template data. The slice template data includes information indicating the hierarchical structure of those elements which is illustrated in FIG. 5.

(113) The inventory template data is, for example, data corresponding to a part of the technology section data and a part of the security section data included in the bundle file. The inventory template data is, for example, template data indicating the logic used by the inventory management module **98**.

(114) The CM template data is, for example, data corresponding to a part of the technology section data and a part of the operation section data included in the bundle file, and is, for example, template data indicating the logic used by the configuration management module **100**.

(115) The monitoring script data is, for example, data corresponding to a part of the operation section data included in the bundle file, and is, for example, data indicating a monitoring script executed by the monitoring management module **102**. As described above, the monitoring script data may include the above-mentioned calculation logic data.

(116) The security script data is, for example, data corresponding to a part of the security section data included in the bundle file, and is, for example, data indicating a script regarding security executed by the security setting module **104**.

(117) The Helm chart data is, for example, data corresponding to a part of the operation section data included in the bundle file, and is data indicating a script template (Helm chart) used by the container management module **106**.

(118) The container image data is, for example, data corresponding to a part of the operation section data included in the bundle file, and is, for example, data of container image of a container included in the functional unit group that achieves the network service. The container image data includes one or a plurality of container images. A container image ID, which is an identifier of the container image, is linked to each of the one or the plurality of container images.

(119) In this embodiment, in response to the reception of a bundle file, the bundle development module **90** determines a bundle ID that corresponds to a data group generated based on the bundle file. A bundle ID is uniquely assigned to each generated data group.

(120) Then, the bundle development module **90** links the product catalog data included in the data group corresponding to the bundle ID to the determined bundle ID, and then transmits the product catalog data to the MPS **30**.

(121) Further, the bundle development module **90** outputs the service catalog data included in the data group to the service management module **92** after linking the service catalog data to the determined bundle ID. Then, the service management module **92** stores the service catalog data in the service catalog storage **94**.

(122) Further, the bundle development module **90** links the inventory template data, the CM template data, the monitoring script data, the security script data, the Helm chart data, and the container image data to the bundle ID corresponding to the data group, and then stores the pieces of data in the inventory management module **98**, the configuration management module **100**, the

monitoring management module **102**, the security setting module **104**, the container management module **106**, and the repository module **108**, respectively.

(123) As described above, in this embodiment, the product catalog data, the service catalog data, the inventory template data, the CM template data, the monitoring script data, the security script data, the Helm chart data, and the container image data become linked to one another by the bundle ID.

(124) Further, in this embodiment, the vendor can easily provide the network service by a simple operation of, for example, specifying a path of the bundle file.

(125) In this embodiment, for example, the product management module **80** receives the product catalog data linked to the bundle ID transmitted from the bundle development module **90**. Then, the product management module **80** stores the received product catalog data in the product catalog storage **82**.

(126) In this embodiment, for example, the product catalog storage **82** stores the product catalog data linked to the bundle ID as described above.

(127) In this embodiment, for example, the purchase management module **84** receives from the purchaser terminal **34** a network service construction request, such as a purchase request for a network service, which is linked to the bundle ID and the service requirement data. A bundle ID linked to a purchase request is hereinafter referred to as “purchase bundle ID,” and service requirement data linked to a purchase request is hereinafter referred to as “purchase service requirement data.”

(128) Then, the purchase management module **84** transmits the purchase service requirement data linked to the purchase bundle ID to the E2EO module **96** in response to the reception of the purchase request described above. The E2EO module **96** receives the purchase service requirement data indicating the service requirements.

(129) In this embodiment, the inventory database **110** is, for example, a database in which inventory information for a plurality of servers **120** managed by the NOS **32** and arranged in the RAN **38** and the core network system **40** is stored.

(130) In this embodiment, for example, the inventory database **110** stores inventory data including the physical inventory data and the logical inventory data. The inventory data indicates the current statuses of the configuration of an element group included in the communication system **1** and the link between the elements. In addition, the inventory data indicates the status of resources managed by the NOS **32** (for example, resource usage status).

(131) FIG. **10** is a diagram for illustrating an example of the data structure of the physical inventory data. The physical inventory data illustrated in FIG. **10** is associated with one server **120**. The physical inventory data illustrated in FIG. **10** includes, for example, a server ID, location data, building data, floor number data, rack data, specification data, network data, an operating container ID list, and a cluster ID.

(132) The server ID included in the physical inventory data is, for example, an identifier of the server **120** associated with the physical inventory data.

(133) The location data included in the physical inventory data is, for example, data indicating the location of the server **120** (for example, the address of the location) associated with the physical inventory data.

(134) The building data included in the physical inventory data is, for example, data indicating a building (for example, a building name) in which the server **120** associated with the physical inventory data is arranged.

(135) The floor number data included in the physical inventory data is, for example, data indicating a floor number at which the server **120** associated with the physical inventory data is arranged.

(136) The rack data included in the physical inventory data is, for example, an identifier of a rack in which the server **120** associated with the physical inventory data is arranged.

(137) The specification data included in the physical inventory data is data indicating the

specifications of the server **120**, such as the number of cores, the memory capacity, and the hard disk capacity, of the server **120** associated with the physical inventory data.

(138) The network data included in the physical inventory data is, for example, data indicating an NIC included in the server **120** associated with the physical inventory data, the number of ports included in the NIC, and the like.

(139) The operating container ID list included in the physical inventory data is, for example, data indicating a list of identifiers (container IDs) of one or a plurality of container instances operating in the server **120** associated with the physical inventory data.

(140) The operating container ID list included in the physical inventory data in this embodiment can be said to be data indicating a part of a current status of hardware on which a software element (container in the above-mentioned example) included in the communication system **1** is operating. The inventory data may include data other than the operating container ID list, which indicates the current status of the hardware on which the software element included in the communication system **1** is operating. For example, the inventory data may indicate the current status of the hardware on which the software element, such as an NS, an NF, a CNFC, a pod, a network slice (NSI), or a network slice subnet instance (NSSI), is operating.

(141) The cluster ID included in the physical inventory data is, for example, an identifier of a cluster (for example, Kubernetes cluster) to which the server **120** associated with the physical inventory data belongs.

(142) In addition, the logical inventory data includes topology data for a plurality of elements included in the communication system **1**, which indicates the current status of such link between the elements as illustrated in FIG. **5**. For example, the logical inventory data includes topology data including an identifier of a certain NS and an identifier of one or a plurality of NFs under the control of the certain NS. In addition, for example, the logical inventory data includes topology data including an identifier of a certain network slice and an identifier of one or a plurality of NFs belonging to the certain network slice.

(143) FIG. **11** is a diagram for schematically illustrating an example of the topology data. The topology data illustrated in FIG. **11** schematically indicates an example of the current statuses of the gNB **56** illustrated in FIG. **3** and the link between the elements (hierarchical structure) between the element group under the control thereof.

(144) FIG. **12** is a diagram for schematically illustrating another example of the topology data. The topology data illustrated in FIG. **12** schematically indicates an example of the current statuses of the network slice **68b** illustrated in FIG. **4** and the elements (for example, NS and NF) belonging to the network slice **68b**.

(145) In FIG. **11** and FIG. **12**, the topology data is represented as data including nodes **130** associated with elements included in the communication system **1** and links **132** each connecting the nodes **130** associated with respective two elements linked to each other. More specifically, in FIG. **11** and FIG. **12**, the topology data is represented as tree structure data in which a hierarchical structure between the elements is represented by a parent-child relationship.

(146) A node **130a** having an identifier of “gNB #1,” which is illustrated in FIG. **11**, is associated with the gNB **56** illustrated in FIG. **3**. A node **130b** having an identifier of “DU #1” is associated with the DU **50** illustrated in FIG. **3**. A node **130c** having an identifier of “CU-CP #1” is associated with the CU-CP **52** illustrated in FIG. **3**. A node **130d** having the identifier of “CU-UP #1” is associated with the CU-UP **54a** illustrated in FIG. **3**. A node **130e** having an identifier of “CU-UP #2” is associated with the CU-UP **54b** illustrated in FIG. **3**. A node **130f** having an identifier of “CU-UP #3” is associated with the CU-UP **54c** illustrated in FIG. **3**. FIG. **11** also shows a node **130g** and a node **130h** corresponding to the CNFCs included in the DU **50**.

(147) FIG. **11** also shows a link **132a** connecting the node **130a** and the node **130b** that is a node **130** being a child of the node **130a**. FIG. **11** also shows a link **132b** connecting the node **130a** and the node **130c** that is a node **130** being a child of the node **130a**. FIG. **11** also shows a link **132c**

connecting the node **130a** and the node **130d** that is a node **130** being a child of the node **130a**. FIG. **11** also shows a link **132d** connecting the node **130a** and the node **130e** that is a node **130** being a child of the node **130a**. FIG. **11** also shows a link **132e** connecting the node **130a** and the node **130f** that is a node **130** being a child of the node **130a**. FIG. **11** also shows a link **132f** connecting the node **130b** and the node **130g** that is a node **130** being a child of the node **130b**. FIG. **11** also shows a link **132g** connecting the node **130b** and the node **130h** that is a node **130** being a child of the node **130b**.

(148) Further, a node **130i** having an identifier of “eMBB #1,” which is illustrated in FIG. **12**, is associated with the network slice **68b** illustrated in FIG. **4**. A node **130j** having the identifier of “gNB #1” is associated with the gNB **56** illustrated in FIG. **3**. A node **130k** having an identifier of “Core #1” is associated with the NS of the core network system **40** belonging to the network slice **68b**. A node **130l** having the identifier of “DU #1” is associated with the DU **50** illustrated in FIG. **3** and FIG. **4**. A node **130m** having the identifier of “CU-UP #2” is associated with the CU-UP **54b** illustrated in FIG. **3** and FIG. **4**. A node **130n** having an identifier of “UPF #2” is associated with the UPF **64b** illustrated in FIG. **3** and FIG. **4**.

(149) FIG. **12** also shows a link **132h** connecting the node **130i** and the node **130j** that is a node **130** being a child of the node **130i**. FIG. **12** also shows a link **132i** connecting the node **130i** and the node **130k** that is a node **130** being a child of the node **130i**. FIG. **12** also shows a link **132j** connecting the node **130j** and the node **130l** that is a node **130** being a child of the node **130j**. FIG. **12** also shows a link **132k** connecting the node **130j** and the node **130m** that is a node **130** being a child of the node **130j**. FIG. **12** also shows a link **132l** connecting the node **130k** and the node **130n** that is a node **130** being a child of the node **130k**.

(150) In addition, the logical inventory data may include NSI data being data indicating attributes such as an identifier of an instance of a network slice and the type of the network slice. In addition, the logical inventory data may include NSSI data being data indicating attributes such as an identifier of an instance of a network slice subnet and the type of the network subnet.

(151) The logical inventory data may also include NS data being data indicating attributes such as an identifier of an instance of an NS and the type of the NS. The logical inventory data may also include NF data indicating attributes such as an identifier of an instance of an NF and the type of the NF. The logical inventory data may also include CNFC data indicating attributes such as the identifier of a CNFC instance and the type of the CNFC. The logical inventory data may also include pod data indicating attributes such as an identifier of an instance of a pod included in the CNFC and the type of the pod. The logical inventory data may also include container data indicating attributes such as a container ID of an instance of a container included in the pod and the type of the container.

(152) With the container ID of the container data included in the logical inventory data and the container ID included in the operating container ID list included in the physical inventory data, an instance of the container and the server **120** on which the instance of the container is operating become linked to each other.

(153) Further, data indicating various attributes such as the host name and the IP address may be set in the above-mentioned data included in the logical inventory data. For example, the container data may include data indicating the IP address of a container corresponding to the container data. Further, for example, the CNFC data may include data indicating the IP address and the host name of a CNFC indicated by the CNFC data.

(154) The logical inventory data may also include data indicating NSSAI including one or a plurality of pieces of S-NSSAI, which is set for each NF.

(155) Further, in this embodiment, at least a part of the inventory data stored in the inventory database **110** is also stored in the active inventory **114**. The inventory data stored in the active inventory **114** and the inventory data stored in the inventory database **110** are basically synchronized with each other. When a process such as registration, update, or deletion of inventory

data occurs, the inventory management module **98** executes the process on the inventory database **110** after executing the process on the active inventory **114**.

(156) Further, the inventory management module **98** can appropriately grasp the resource status in cooperation with the container management module **106**. Then, the inventory management module **98** appropriately updates the inventory data stored in the active inventory **114** and the inventory database **110** based on the latest resource status.

(157) In this embodiment, the E2EO module **96** and the inventory management module **98** identify the configuration of the element group for achieving a network service that satisfies the service requirements indicated by the service requirement data based on, for example, the inventory data and the service requirement data received from the purchase management module **84**.

(158) For example, the E2EO module **96** acquires, from the service catalog storage **94**, service catalog data corresponding to a purchase bundle ID linked to the purchase service requirement data received from the purchase management module **84**. The E2EO module **96** then executes a workflow script indicated by the service catalog data.

(159) [Construction of Functional Unit Group Corresponding to Purchase of Network Service]

(160) The E2EO module **96** and the inventory management module **98** determine the configuration of a network service to be purchased based on the purchase service requirement data received from the purchase management module **84**, the service catalog data linked to the purchase bundle ID, the inventory template data linked to the purchase bundle ID, and the inventory data.

(161) In this case, for example, the E2EO module **96** may identify the respective types of functional unit groups that achieve the network service to be purchased and the number of functional units for each type, based on the purchased service requirement data and the requirement configuration correspondence data included in the service catalog data.

(162) Further, based on the purchase service requirement data, the E2EO module **96** may identify the location at which each of the functional units included in the constructed functional unit group is constructed. For example, the E2EO module **96** may determine the location of each functional unit included in the constructed functional unit group based on the target area indicated by the purchase service requirement data and the requirement configuration correspondence data included in the service catalog data. A different location may be determined for each functional unit.

(163) Further, based on the purchase service requirement data, the E2EO module **96** may identify, for each of a plurality of locations, the type and the number of functional units constructed at the location. In this case, in accordance with the location that is identified based on the purchase service requirement data, the E2EO module **96** may determine the number of functional units for each type that are constructed at the location. Further, the E2EO module **96** may determine the number of functional units for each type that are constructed for each location based on a weight set for each location identified based on the purchase service requirement data.

(164) An example of the weight set for each location is the population of an area covered by one or a plurality of cells under the control of the data center linked to the location. That is, in determining the number of functional units of each type, which are constructed for each location, weighting may be performed based on the population of the covered area. For example, as the population of the area covered by a location becomes larger, more functional units may be allocated at the location.

(165) In this case, the E2EO module **96** may store assumed busy level data for each of a plurality of locations, which indicates the population of the area covered by one or a plurality of cells under the control of the data center linked to the location. The value of the assumed busy level data is an example of the weight set for each location described above. The assumed busy level data for the data center of the core network system **40** may indicate, for example, the population of the area covered by the cells of the RAN **38** communicating to/from the core network system **40**. For example, as the population at a location which is indicated by the assumed busy level data becomes larger, more functional units may be deployed at the location.

(166) Then, the E2EO module **96** and the inventory management module **98** register the inventory data associated with the newly constructed element group in the active inventory **114** and the inventory database **110** based on the number and the location of functional units, which are determined as described above, and the inventory data.

(167) In this case, the inventory management module **98** may determine a host name and an IP address that are to be allocated to the newly constructed element group based on the inventory data. In this case, for example, a host name and an IP address may be determined so as not to overlap with the already used host name and the already used IP address. Then, the inventory data in which the determined host name and IP address are set may be registered in the active inventory **114** and the inventory database **110**.

(168) In addition, the E2EO module **96** and the inventory management module **98** may identify the existing element group to which a new setting is to be input, based on the number of functional units and the location, which are determined as described above, and the inventory data. Then, the inventory data indicating the new setting to be input may be registered in the active inventory **114** and the inventory database **110**.

(169) A planned attribute is set for the inventory data registered in response to the reception of the service requirement data as described above. The inventory data for which the planned attribute is set is hereinafter referred to as “planned data.” The planned data registered in this manner is linked to an inventory key being an identifier of the planned data, the above-mentioned purchase bundle ID, and a user ID being an identifier of the purchaser (user) who has made the purchase request.

(170) The E2EO module **96** and the inventory management module **98** may determine whether to construct a new element or to use an existing element as an element included in the purchased NS based on the current status (for example, operation status or availability status) of the element group indicated by the inventory data. Then, the E2EO module **96** and the inventory management module **98** may register, based on the determination, in the active inventory **114** and the inventory database **110**, the planned data associated with the newly constructed element group and the planned data indicating a new setting to be input to the existing elements.

(171) In this embodiment, based on, for example, the configuration of the element group identified as described above and template data in which the configuration is acceptable as a parameter, the E2EO module **96** and the configuration management module **100** identify a construction procedure of the element group. The construction procedure includes, for example, a procedure of container configuration management such as deploying a container and setting the deployed container and a container related to the deployed container. This process is executed, for example, with the execution of the workflow script by the E2EO module **96** as a trigger.

(172) Then, the E2EO module **96**, the configuration management module **100**, and the container management module **106** construct an element group that achieves the network service based on the configuration identified as described above. The E2EO module **96**, the configuration management module **100**, and the container management module **106** construct an element group by executing, for example, the identified construction procedure. This process is also executed, for example, with the execution of the workflow script by the E2EO module **96** as a trigger. Each of the elements included in the element group may be constructed at a location identified for the element. Further, for example, the number of element groups identified based on the purchase service requirement data may be constructed. Further, for example, for each of a plurality of locations, an identified number of elements of a type identified for the location may be constructed.

(173) The configuration management module **100** and the bare metal management module **112** secure, for example, a hardware resource (for example, the server **120** in this case) in which a new element group is deployed.

(174) Further, the configuration management module **100** and the bare metal management module **112** perform a system software setup in accordance with a specific type of element on an unused hardware resource. In this embodiment, for example, the configuration management module **100** or

the bare metal management module **112** stores a script (for example, an Ansible script) for performing a setup for the above-mentioned specific type of element. The script describes, for example, a procedure of installing a host OS, which have a specific type or a specific version, serving as a platform of a container execution environment, a procedure of setting a kernel of the host OS, and a procedure of setting a basic input output system (BIOS). Then, by the execution of the script by the bare metal management module **112**, a system software setup in accordance with the specific type of element is performed on a free server. For example, the setup of the host OS and the BIOS of the container execution environment is performed on the free server.

(175) When the hardware resource for deploying a new element group is secured, the E2EO module **96** instructs the container management module **106** to deploy the new element group, for example, based on the above-mentioned planned data and service template data linked to the purchase bundle ID stored in the E2EO module **96**. The service template data can accept a part or all of the planned data as a parameter.

(176) An NS descriptor (NSD) is an example of the above-mentioned service template data. The NSD is associated with a network service, and indicates, for example, the types of a plurality of functional units (for example, a plurality of CNFs) included in the network service. The NSD may indicate the number of CNFs or other functional units included in the network service for each type thereof. The NSD may also indicate a file name of a CNFD described later, which relates to the CNF included in the network service.

(177) Further, a CNF descriptor (CNFD) is an example of the above-mentioned service template data. The E2EO module **96** generates, for example, a day 0 parameter (CNF instance) based on the planned data and the CNFD. For example, the day 0 parameter in which a host name and the value of an IP address of the CNFD are set is generated.

(178) The CNFD may indicate computer resources (for example, CPU, memory, and hard disk drive) required by the CNF. For example, the CNFD may also indicate, for each of a plurality of containers included in the CNF, computer resources (for example, CPU, memory, and hard disk drive) required by the container.

(179) The CNFD may include a template associated with each of a plurality of deployment flavors. Then, for example, the E2EO module **96** may generate the day 0 parameter based on a template corresponding to a deployment flavor in accordance with the purchase service requirement data.

(180) The E2EO module **96** may identify the location of the output destination of the day 0 parameter. For example, one or a plurality of container management modules **106** that serve as output destinations of the day 0 parameter may be identified. For example, a container management module **106** associated with a server **120** arranged at the location of an element indicated by the planned data may be identified. Then, a day 0 parameter that is output to each of the identified locations may be generated. For example, a day 0 parameter that is output to each of one or a plurality of container management modules **106** that serve as the output destinations may be generated.

(181) Then, the E2EO module **96** outputs each of the generated one or plurality of day 0 parameters to a container management module **106** serving as the location of the output destination of the day 0 parameter. A purchase bundle ID is linked to the day 0 parameter.

(182) Then, the container management module **106** deploys a new element group based on the received day 0 parameter. The container management module **106** identifies a container image to be deployed and the server **120** in which the container is deployed, for example, based on Helm chart data associated with the purchase bundle ID and on the received day 0 parameter. Then, the container management module **106** acquires the container image from the repository module **108** and deploys a container corresponding to the container image in the identified server **120**. For example, a manifest file is generated based on the Helm chart data associated with the purchase bundle ID and on the received day 0 parameter. Then, the deployment of a container is executed through use of the manifest file.

(183) In this embodiment, each of the plurality of servers **120** may be allocated to a resource pool. In this case, the container management module **106** may identify a container image to be deployed and a resource pool in which the container is deployed, for example, based on Helm chart data associated with the purchase bundle ID and on the received day 0 parameter. Then, the container management module **106** may acquire the container image from the repository module **108** and deploy a container corresponding to the container image in the identified resource pool.

(184) Further, the configuration management module **100** generates planned CM data including a day 1 parameter, for example, based on the above-mentioned planned data and CM template data stored in the configuration management module **100** and linked to the purchase bundle ID. The CM template data can accept a part or all of the planned data as a parameter.

(185) The day 1 parameter indicates, for example, a configuration management procedure, for example, the settings of a deployed element group and at least one element related to the element group (for example, an element communicating to/from the deployed element group). A day 1 parameter relating to the RAN **38** indicates, for example, radio field intensity, the direction and angle of the antenna **16**, a serial number, and the like. A day 1 parameter relating to a UPF indicates, for example, information indicating an opposite node.

(186) Then, the configuration management module **100** executes configuration management, for example, the setting of the element group, based on the day 1 parameter included in the generated planned CM data. Those processes are executed, for example, with the execution of the workflow script by the E2EO module **96** as a trigger. In this embodiment, for example, configuration management is executed for a newly constructed element group or an existing element to which a new setting is to be input.

(187) Then, the E2EO module **96** executes, for example, instantiation of a network slice pertaining to a network service to be purchased, based on the above-mentioned planned data and slice template data linked to the purchase bundle ID stored in the E2EO module **96**. The slice template data can accept a part or all of the planned data as a parameter. This process is executed, for example, with the execution of the workflow script by the E2EO module **96** as a trigger.

(188) The E2EO module **96** includes a network slice management function (NSMF) and a network slice sub-network management function (NSSMF) described in the 3GPP specification “TS28 533.” The NSMF is a function for generating and managing network slices, and provides NSI management. The NSSMF is a function for generating and managing network slice subnets forming a part of a network slice, and provides NSSI management.

(189) The E2EO module **96** may output a configuration management instruction related to the instantiation of the network slice to the configuration management module **100**. Then, the configuration management module **100** may execute configuration management, for example, settings in accordance with the configuration management instruction.

(190) For example, the configuration management module **100** constructs the NSI and the NSSI on each server **120** by using known segment routing technology (for example, SRv6 (segment routing IPv6)) based on the setting information passed from the E2EO module **96**. For example, the configuration management module **100** may generate an NSI and NSSI extending over a plurality of NFs to be set by issuing, to the plurality of NFs to be set, a command to set a common virtual local area network (VLAN) and a command to assign a bandwidth and a priority indicated by the setting information to the VLAN.

(191) Further, for example, the configuration management module **100** may execute configuration management regarding new element groups when the deployment of the new element groups is completed, and then execute configuration management related to the instantiation of the network slice.

(192) As another example, the configuration management module **100** may update a once-generated day 1 parameter based on the configuration management instruction received from the E2EO module **96**. Then, the configuration management module **100** may collectively perform the

configuration management related to the new element groups and the instantiation of the network slice.

(193) In this embodiment, for example, the monitoring management module **102** identifies a monitoring policy indicated by the purchase service requirement data based on the above-mentioned planned data and monitoring script data linked to the purchase bundle ID stored in the monitoring management module **102**. Then, the monitoring management module **102** executes a monitoring setting in accordance with the identified monitoring policy.

(194) Then, in accordance with the identified monitoring policy, the monitoring management module **102** monitors an element group which is constructed. For example, monitoring of a monitoring target indicated by the purchase service requirement data may be executed at a monitoring interval indicated by the purchase service requirement data. This process is executed, for example, with the execution of the workflow script by the E2EO module **96** as a trigger.

(195) In this embodiment, the monitoring management module **102** executes monitoring at various levels, for example, a slice level, an NS level, an NF level, a CNFC level, and a level of hardware, for example, the server **120**.

(196) For example, the monitoring management module **102** may set a module for outputting metric data in the hardware, for example, the server **120**, or an element illustrated in FIG. 5 so that monitoring can be performed at the various levels described above. In this case, for example, the NF may output the metric data indicating a metric that can be measured by the NF to the monitoring management module **102**. Further, the server **120** may output the metric data indicating a metric relating to the hardware that can be measured by the server **120** to the monitoring management module **102**.

(197) In addition, for example, the monitoring management module **102** may deploy, in the server **120**, a sidecar container for aggregating the metric data indicating the metrics output from a plurality of containers in units of CNFCs (microservices). This sidecar container may include an agent called “exporter.” The monitoring management module **102** may repeatedly execute a process for acquiring the metric data aggregated in units of microservices from the sidecar container, at monitoring intervals indicated by the purchase service requirement data through use of the mechanism of Prometheus.

(198) Further, the elements such as the network slice, the NS, the NF, the CNFC that are included in the communication system **1** and the hardware, for example, the server **120**, notify the monitoring management module **102** of various alerts (for example, notify the monitoring management module **102** of an alert with the occurrence of a failure as a trigger).

(199) For example, in this embodiment, the security setting module **104** executes a security setting, for example, a password setting, in accordance with the value of the purchase service requirement data based on, for example, the above-mentioned planned data and the security script data stored in the security setting module **104** and linked to the purchase bundle ID.

(200) When the inventory management module **98** ends a series of processes including the construction of the element group and the configuration management in the above-mentioned manner, the inventory management module **98** updates the inventory data so as to indicate the configuration of the constructed element group. In this case, for example, the inventory management module **98** cancels the setting of the planned attribute regarding the planned data. In this manner, the inventory data, which has played a role of the planned data, has the role shifted from the role of the planned data to a role of the inventory data indicating the current status of the configuration of the element group actually constructed in the communication system **1** and the current status of the link between the elements.

(201) [Monitoring]

(202) In this case, the monitoring executed by the monitoring management module **102** is further described.

(203) As described above, in this embodiment, the inventory data stored in the inventory database

110 and the active inventory **114** indicates the current status of the configuration of the element group included in the communication system **1** and the current status of the link between the elements included in the communication system **1**.

(204) Further, as described above, the monitoring management module **102** acquires metric data indicating the metric measured by an element from the element included in the communication system **1**.

(205) Then, in this embodiment, the monitoring management module **102** executes the monitoring script included in the monitoring script data stored in the monitoring management module **102**, to thereby calculate a desired performance index value such as the KPI of a specific type of element (for example, slice, NS, or NF) based on the acquired metric data.

(206) In addition, in this embodiment, as described above, the monitoring script data includes the calculation logic data. In this embodiment, a part of the calculation logic indicated by the calculation logic data is a parameter (variable), and this parameter is substituted by a value determined based on the inventory data.

(207) Now, as a first example, consideration is given to a case in which the calculation logic data indicates a calculation logic for calculating, for a specific type of NS (for example, gNB), an average value of the availability of all NFs (parameter x1) under the control of the NS as the availability of the NS.

(208) In this case, it is assumed that the monitoring management module **102** has acquired, for every NF, the metric data indicating the availability of the NF.

(209) In this case, the monitoring management module **102** refers to the inventory data to identify one or a plurality of NSes of the specific type indicated by the inventory data.

(210) In this case, one of the NSes identified in this manner is referred to as “NS of interest.” In this case, for example, it is assumed that the gNB **56** has been identified as the NS of interest.

(211) Then, the monitoring management module **102** identifies all the NFs under the control of the NS of interest based on the inventory data. In this case, for example, the monitoring management module **102** identifies the CU-CP **52**, the CU-CP **54a**, the CU-CP **54b**, and the CU-CP **54c** as the NFs under the control of the NS of interest based on the topology data illustrated in FIG. **11**.

(212) Then, the monitoring management module **102** substitutes one or a plurality of NFs identified in this manner as the value of the parameter x1 of the above-mentioned calculation logic data.

(213) Then, the monitoring management module **102** calculates the value of the availability of the NS of interest in accordance with the calculation logic indicated by the calculation logic data. In this case, for example, the monitoring management module **102** identifies the availability of one or a plurality of NFs set as the value of the parameter x1. Then, the monitoring management module **102** calculates an average value of the identified availability as the value of the availability of the NS of interest.

(214) In this embodiment, for example, the above-mentioned process is executed for all the identified NSes of the specific type.

(215) Now, as a second example, consideration is given to a case in which the calculation logic data indicates a calculation logic for calculating, for a specific type of network slice, an average value of the availability of all NFs (parameter x2) belonging to the network slice as the availability of the network slice.

(216) In this case, it is assumed that the monitoring management module **102** has acquired the metric data indicating the availability of every NF. In this case, it is conceivable that the NF belongs to a plurality of network slices. In this case, the monitoring management module **102** acquires the metric data indicating the availability of every NF for each network slice.

(217) In this case, the monitoring management module **102** refers to the inventory data to identify one or a plurality of network slices of the specific type indicated by the inventory data. In this case, one of the network slices identified in this manner is referred to as “network slice of interest.” In this case, for example, it is assumed that the network slice **68b** has been identified as the network

slice of interest.

(218) Then, the monitoring management module **102** identifies all the NFs belonging to the network slice of interest based on the inventory data. In this case, for example, the monitoring management module **102** identifies the DU **50**, the CU-UP **54b**, and the UPF **64b** as the NFs belonging to the network slice of interest based on the topology data illustrated in FIG. **12**.

(219) Then, the monitoring management module **102** substitutes one or a plurality of NFs identified in this manner as the value of the parameter x2 of the above-mentioned calculation logic data.

(220) Then, the monitoring management module **102** calculates the value of the availability of the network slice of interest in accordance with the calculation logic indicated by the calculation logic data. In this case, for example, the monitoring management module **102** identifies the availability of one or a plurality of NFs set as the value of the parameter x2 in regard to the network slice of interest. As illustrated in FIG. **4**, the DU **50**, the CU-UP **54b**, and the UPF **64b** all belong to a plurality of network slices, but in this case, the availability regarding the network slice **68b** is identified. Then, the monitoring management module **102** calculates an average value of the identified availability as the value of the availability of the network slice of interest.

(221) In this embodiment, for example, the above-mentioned process is executed for all the identified network slices of the specific type.

(222) In the second example, the monitoring management module **102** may identify all the NSes belonging to the network slice for each network slice of interest. For example, the NSes corresponding to the node **130j** and the node **130k** may be identified.

(223) Then, the monitoring management module **102** may identify, for each of the identified NSes, all the NFs belonging to the network slice under the control of the NS. In this case, for example, the NFs corresponding to the node **130l** and the node **130m** may be identified as the NFs under the control of the node **130j**. Meanwhile, the NF corresponding to the node **130n** may be identified as the NF under the control of the node **130k**.

(224) Then, the monitoring management module **102** may calculate, for each of the identified NSes, an average value of the availability of all the NFs belonging to the network slice under the control of the NS in regard to the network slice of interest as the value of the availability of the NS. For example, an average value of the availability of the NFs respectively corresponding to the node **130l** and the node **130m** in regard to the network slice corresponding to the node **130i** may be calculated as the value of the availability of the NS corresponding to the node **130j** in regard to the network slice corresponding to the node **130i**. Meanwhile, the value of the availability of the NF corresponding to the node **130n** in regard to the network slice corresponding to the node **130i** may be calculated as the value of the availability of the NS corresponding to the node **130k** in regard to the network slice corresponding to the node **130i**.

(225) Then, the monitoring management module **102** may calculate an average value of the values of the availability calculated for the respective NSes as the value of the availability of the network slice of interest. For example, an average value of the value of the availability of the NS corresponding to the node **130j** in regard to the network slice corresponding to the node **130i** and the value of the availability of the NS corresponding to the node **130k** in regard to the network slice corresponding to the node **130i** may be calculated as the value of the availability of the network slice corresponding to the node **130i**.

(226) In this case, for example, the calculation logic data may indicate a calculation logic for calculating a weighted average value of the values of the availability calculated for the respective NSes as the value of the availability of the network slice.

(227) Then, the monitoring management module **102** may calculate the weighted average value of the values of the availability calculated for the respective NSes as the value of the availability of the network slice.

(228) For example, the calculation logic data may indicate a calculation logic for calculating “(p1×99+p2)/100” as the availability of the network slice. In this case, p1 represents the average

value of the availability of the NFs included in the RAN **38**, and $p2$ represents the average value of the availability of the NFs included in the core network system **40**.

(229) Then, in this case, the monitoring management module **102** may calculate the value of the availability of the network slice of interest by applying the above-mentioned calculation logic. For example, the value of the availability of the NS corresponding to the node **130j** in regard to the network slice corresponding to the node **130i** is represented by $p1$, and the value of the availability of the NS corresponding to the node **130k** in regard to the network slice corresponding to the node **130i** is represented by $p2$. In this case, the value calculated by the mathematical expression " $(p1 \times 99 + p2) / 100$ " may be calculated as the value of the availability of the network slice corresponding to the node **130i**.

(230) In this embodiment, as described above, the NOS **32** may store the inventory data indicating the current status of the link between the elements included in the communication system **1**.

(231) Then, the NOS **32** may store a calculation logic data indicating a calculation logic for calculating the performance index value of a specific type of element based on the performance index value of the element group linked to the element. The calculation logic data may indicate a calculation logic for calculating the performance index value of a specific type of element based on the performance index value of the element group of the child of the element. As described in the first example, the calculation logic data may also indicate a calculation logic for calculating the performance index value of a specific type of NS based on the performance index value of an NF group under the control of the NS. As described in the second example, the calculation logic data may also indicate a calculation logic for calculating the performance index value of a specific type of network slice based on the performance index value of an NF group belonging to the network slice.

(232) Then, the monitoring management module **102** may identify the above-mentioned specific type of element included in the communication system **1** based on the inventory data and the calculation logic data. For example, as described in the first example, the monitoring management module **102** may identify a specific type of NS included in the communication system **1** based on the inventory data and the calculation logic data. As described in the second example, the monitoring management module **102** may also identify a specific type of network slice included in the communication system **1** based on the inventory data and the calculation logic data.

(233) Then, the monitoring management module **102** may identify the performance index value of each of the plurality of elements included in the element group linked to the identified specific type of element. For example, as described in the first example, the monitoring management module **102** may identify the performance index value of each of the plurality of NFs included in an NF group under the control of the identified specific type of NS. As described in the second example, the monitoring management module **102** may also identify the performance index value of each of the plurality of NFs included in an NF group belonging to the identified specific type of network slice in regard to the network slice.

(234) Then, the monitoring management module **102** may calculate the performance index value of a specific type of element by applying the calculation logic indicated by the calculation logic data to the performance index value identified for each of the plurality of elements. For example, as described in the first example, the monitoring management module **102** may calculate the performance index value of a specific type of NS by applying the calculation logic indicated by the calculation logic data to the performance index value identified for each of the plurality of NFs. As described in the second example, the monitoring management module **102** may also calculate the performance index value of a specific type of network slice by applying the calculation logic indicated by the calculation logic data to the performance index value identified for each of the plurality of NFs. In this case, the type of the performance index represented by the performance index value identified for each of the plurality of elements and the type of the performance index represented by the performance index value calculated by applying the calculation logic indicated

by the calculation logic data to the performance index value may be the same or different from each other.

(235) Further, in this embodiment, as described above, the NOS **32** may store the inventory data indicating the current status of the configuration of the element group included in the communication system **1**.

(236) The NOS **32** may also store the calculation logic data indicating a condition to be satisfied by an element used for calculating the desired performance index value, a monitoring item for the element, and a calculation logic for calculating the desired performance index value.

(237) For example, in regard to the calculation logic data described in the first example, the condition that the NF is under the control of a specific type of NS corresponds to the “condition to be satisfied by an element used for calculating the desired performance index value.” The availability of the NF corresponds to the “monitoring item for the element.” The calculation logic for calculating the average value of the availability of all the NFs under the control of the NS as the availability of the NS corresponds to the “calculation logic for calculating the desired performance index value.”

(238) Meanwhile, in regard to the calculation logic data described in the second example, the condition that the NF belongs to a specific type of network slice corresponds to the “condition to be satisfied by an element used for calculating the desired performance index value.” The availability of the NF in regard to the network slice corresponds to the “monitoring item for the element.” The calculation logic for calculating the average value of the availability of all the NFs belonging to the network slice as the availability of the network slice corresponds to the “calculation logic for calculating the desired performance index value.”

(239) Then, the monitoring management module **102** may identify an element that satisfies the above-mentioned condition from the element group indicated by the inventory data based on the inventory data and the calculation logic data.

(240) Then, the monitoring management module **102** may calculate a desired performance index value by applying the calculation logic indicated by the calculation logic data to the performance index value of the above-mentioned monitoring item for the identified element.

(241) The element for which the calculation logic for calculating the performance index value is indicated by the calculation logic data is not limited to the network slice (NSI) or the network service (NS). For example, a calculation logic for calculating the performance index value of the NSSI or the NF may be indicated. For example, the calculation logic data may indicate a calculation logic for calculating a total value of throughputs of all the CNFCs under the control of a specific type of NF as a throughput of the NF. In this case, a condition that the CNFCs are under the control of the NF corresponds to the above-mentioned “condition to be satisfied by an element used for calculating the desired performance index value.” In addition, the throughput corresponds to the above-mentioned “monitoring item for the element.”

(242) In this embodiment, for example, the monitoring management module **102** also notifies the E2EO module **96** of the value of the KPI calculated as described above. Then, the E2EO module **96** executes the workflow script associated with the condition when a predetermined condition that, for example, the value of the KPI exceeds a threshold value defined in advance or the value of the KPI is smaller than a threshold value defined in advance, is satisfied. In this manner, for example, scale-out corresponding to the condition is executed. In this embodiment, the scale-out that cannot be handled by Kubernetes of the container management module **106** can be executed. The information of the threshold value may be included in technology service catalog data of a bundle file. As described above, the information of the threshold value may also be included in the service catalog data.

(243) In the scale-out, the same process as a process performed when a new element is constructed is executed. For example, the inventory management module **98** updates the inventory data stored in the active inventory **114** and the inventory database **110** so as to conform to changes in current

statuses of the configuration of the element group and the link between the elements due to the scale-out.

(244) For example, it is assumed that the scale-out has been executed, to thereby add the CU-UP **54** under the control of the gNB **56** and update the topology data illustrated in FIG. **11** to topology data illustrated in FIG. **13**. In the topology data illustrated in FIG. **13**, a node **130o** having an identifier of “CU-UP #101” has been added. In addition, a link **132m** connecting the node **130a** and the node **130o** to each other has been added.

(245) Further, for example, it is assumed that the scale-out has been executed, to thereby add the UPF **64** belonging to the network slice **68b** and update the topology data illustrated in FIG. **12** to topology data illustrated in FIG. **14**. In the topology data illustrated in FIG. **14**, a node **130p** having an identifier of “UPF #101” has been added. In addition, a link **132n** connecting the node **130k** and the node **130p** to each other has been added.

(246) Even when such a topology change as described above occurs, in this embodiment, only the value substituted into the parameter of the above-mentioned calculation logic data is changed, and it is not required to change the calculation logic itself indicated by the calculation logic data.

(247) For example, the values substituted as the value of the above-mentioned parameter x1 are only changed from the NFs corresponding to the node **130b**, the node **130c**, the node **130d**, the node **130e**, and the node **130f** to the NFs corresponding to the node **130b**, the node **130c**, the node **130d**, the node **130e**, the node **130f**, and the node **130o**. Further, the values substituted as the value of the above-mentioned parameter x2 are only changed from the NFs corresponding to the node **130l**, the node **130m**, and the node **130n** to the NFs corresponding to the node **130l**, the node **130m**, the node **130n**, and the node **130p**.

(248) Accordingly, it is not required to change monitoring settings, for example, change targets to be aggregated, each time there is a change in current status of the link between elements, such as addition or deletion of a functional unit under the control of the network service. In addition, it is not required to set the calculation logic for calculating the performance index value to be monitored in accordance with the current status of a constructed functional block group.

(249) In this manner, according to this embodiment, it is possible to reduce time and labor required in the operation of the communication system **1**.

(250) The topology change in this embodiment is not limited to the scale-out. For example, scale-in or relocation (replacement) may be handled as the topology change in this embodiment.

(251) Further, it is assumed that an NF belonging to a specific network slice has been added by the scale-out in this embodiment. In this case, an NF added in this manner is not only added to an element substituted into the parameter in the calculation of the desired performance index value for an NS having the NF under the control thereof but also added to an element substituted into the parameter in the calculation of the desired performance index value for a network slice to which the NF belongs. That is, the addition of such an NF affects both the calculation of the desired performance index value for the NS having the NF under the control thereof and the calculation of the desired performance index value for the network slice to which the NF belongs.

(252) [Execution of Action]

(253) Now, an example of execution of an action such as the execution of the workflow script by the E2EO module **96** is further described.

(254) FIG. **15** is a diagram for illustrating an example of a configuration of the E2EO module **96**. As illustrated in FIG. **15**, the E2EO module **96** includes a policy manager module **140**, a resource manager module **142**, a state machine module **144**, and a workflow module **146** in addition to the above-mentioned active inventory **114**.

(255) The policy manager module **140** receives a notification of the performance index value such as the value of the KPI described above from the monitoring management module **102**. As described above, the performance index value is calculated by the monitoring management module **102** based on, for example, the metric data acquired by the monitoring management module **102**.

Then, the policy manager module **140** holds the data indicating the performance index value indicated by the received notification.

(256) Further, in this embodiment, when the monitoring management module **102** receives a notification of an alert from each of the elements (software elements) such as the network slice, the NS, the NF, and the CNFC that are included in the communication system **1** and the hardware, for example, the server **120** in such a manner as described above, the monitoring management module **102** transmits the notification to the policy manager module **140**. Then, the policy manager module **140** receives the notification of the alert transmitted from the monitoring management module **102**. Then, the policy manager module **140** holds the data indicating the notification of the alert.

(257) The resource manager module **142** manages hardware resource, software resource, and other various resources included in the communication system **1**.

(258) The resource manager module **142** monitors, for example, usage statuses and availability statuses of resources including a core, a memory, and a hard disk drive that are mounted to the server. Then, the inventory data stored in the inventory database **110** and the active inventory **114** is appropriately updated based on results of the monitoring. In this manner, the inventory data indicates the latest statuses (usage status and availability status) of the resources managed by the NOS **32**.

(259) The resource manager module **142** may acquire metric data output from the monitoring management module **102**, the metric data indicating, for example, the usage status and the availability status of the resources including the core, the memory, and the hard disk drive that are mounted to the server. Then, the resource manager module **142** may appropriately update the inventory data stored in the inventory database **110** and the active inventory **114** based on the metric data acquired from the monitoring management module **102**.

(260) The state machine module **144** stores a state machine associated with an element included in the communication system **1**. The state machine manages a state of the element associated with the state machine.

(261) The workflow module **146** stores the above-mentioned workflow script.

(262) The policy manager module **140** determines, for example, to execute an action such as scale-out on a specific element included in the communication system **1** based on a performance index value or an alert received from the monitoring management module **102**. Then, the policy manager module **140** outputs an execution instruction for an action to the state machine corresponding to an element on which the execution of the action has been determined. Then, the state machine starts to execute the workflow associated with the action on the element.

(263) As described above, the inventory data in this embodiment may indicate the current statuses of the links between the elements included in the communication system **1**. The inventory data may also indicate the current statuses of the software elements included in the communication system **1** and the hardware on which the software element is operating.

(264) Then, in this embodiment, the policy manager module **140** stores action data indicating actions, for example, healing, scale-out, and relocation (replacement) to be executed on the elements included in the communication system **1**. This action data may be, for example, an identifier (ID) of the above-mentioned workflow.

(265) Then, it is assumed that the action data described below is linked to condition data indicating a condition relating to the operation status of a specific type of element.

(266) Then, in this embodiment, for example, the policy manager module **140** identifies a current operation status (for example, the above-mentioned performance index value or the status indicated by the received alert) of the specific type of element included in the communication system **1**.

(267) Then, when the operation status identified for the specific type of element satisfies the condition linked to the action data, the policy manager module **140** instructs the state machine associated with the specific type of element to execute an action indicated by the action data on the specific type of element. Then, the state machine executes the action.

(268) Examples of the condition indicated by the condition data linked to the action data indicating the scale-out of a specific type of NS include “the availability of gNB is equal to or smaller than a predetermined value.” In this case, for example, when the policy manager module **140** identifies that the availability of the gNB **56** illustrated in FIG. 3 is equal to or smaller than the predetermined value, the scale-out of the gNB **56** is executed.

(269) Further, examples of the condition indicated by the condition data linked to the action data indicating the scale-out of a specific type of NF include “the availability of UPF is equal to or smaller than a predetermined value.” In this case, for example, when the policy manager module **140** identifies that the availability of the UPF **64a** illustrated in FIG. 3 is equal to or smaller than the predetermined value, the scale-out of the UPF **64a** is executed.

(270) Further, examples of the condition indicated by the condition data linked to the action data indicating the scale-out of a specific type of network slice include “the availability of network slice of an eMBB is equal to or smaller than a predetermined value.” In this case, for example, when the policy manager module **140** identifies that the availability of the network slice **68b** illustrated in FIG. 4 is equal to or smaller than the predetermined value, the scale-out of the network slice **68b** is executed.

(271) In this embodiment, the action data may be linked to condition data indicating a condition relating to a combination of the operation status of a specific type of element and the operation status of another type of element linked to the specific type of element. It is indicated in, for example, the inventory data which element is linked to the specific type of element as the another type of element.

(272) Then, the policy manager module **140** may identify, based on the current status indicated by the inventory data, a combination of the current operation status of a specific type of element included in the communication system **1** and the current operation status of another type of element linked to the specific type of element.

(273) Then, when the identified combination satisfies the condition linked to the action data, the E2EO module **96** may execute the action indicated by the action data on the specific type of element. For example, when the identified combination satisfies the condition linked to the action data, the policy manager module **140** may instruct the state machine associated with the element to execute the action indicated by the action data on the specific type of element. Then, the state machine may execute the action.

(274) For example, the action data may be linked to condition data indicating a condition relating to a combination of the operation status of a specific type of network service (NS) and the operation status of each of a plurality of functional units (NF) under the control of the specific type of NS.

(275) Then, the policy manager module **140** may identify, based on the current status indicated by the inventory data, a combination of the current operation status of a specific type of NS included in the communication system **1** and the current operation status of each of a plurality of NFs under the control of the NS.

(276) Then, when the identified combination satisfies the condition linked to the action data, the policy manager module **140** may instruct the state machine associated with the NS to execute the action indicated by the action data on the specific type of NS. Then, the state machine may execute the action.

(277) In this case, examples of the condition indicated by the condition data linked to the action data indicating the relocation (replacement) of a specific type of NS include “the availability of a gNB is smaller than a predetermined value a1 and the availability of all the NFs under the control of the gNB is equal to or larger than a predetermined value a2.” It is presumed that this condition is satisfied when hardware on which NFs under the control of an NS are operating has deteriorated on the whole. Accordingly, in general, it is desired that the relocation (replacement) of the NS be executed when this condition is satisfied.

(278) Then, in this case, for example, the policy manager module **140** may identify based on the inventory data that the DU **50**, the CU-CP **52**, the CU-UP **54a**, the CU-UP **54b**, and the CU-UP **54c** are under the control of the gNB **56** illustrated in FIG. **3**. Then, the policy manager module **140** may identify the operation status (for example, value of the availability in this case) of the gNB **56**. The policy manager module **140** may also identify the operation status (for example, value of the availability in this case) of each of the DU **50**, the CU-CP **52**, the CU-UP **54a**, the CU-UP **54b**, and the CU-UP **54c**.

(279) Then, the policy manager module **140** may determine whether or not the value of the availability of the gNB **56** is smaller than the predetermined value a1. In this case, for example, it is assumed that the value of the availability of the gNB **56** is smaller than the predetermined value a1.

(280) In this case, the policy manager module **140** may determine whether or not the values of the availability of the DU **50**, the CU-CP **52**, the CU-UP **54a**, the CU-UP **54b**, and the CU-UP **54c** are all equal to or larger than the predetermined value a2.

(281) Then, when the values of the availability of the DU **50**, the CU-CP **52**, the CU-UP **54a**, the CU-UP **54b**, and the CU-UP **54c** are all equal to or larger than the predetermined value a2, the relocation (replacement) of the gNB **56** may be executed. Specifically, for example, the gNB **56** may be executed on another server, or a server on which the gNB **56** is currently operating may be separated from a cluster to which the server belongs.

(282) Then, when at least one of the values of the availability of the DU **50**, the CU-CP **52**, the CU-UP **54a**, the CU-UP **54b**, and the CU-UP **54c** is smaller than the predetermined value a2, the relocation (replacement) of the gNB **56** may be inhibited from being executed.

(283) Further, for example, the action data may be linked to condition data indicating a condition relating to a combination of the operation status of a specific type of functional unit (NF) and the operation status of a network service (NS) having the specific type of NF under the control thereof.

(284) Then, the policy manager module **140** may identify, based on the current status indicated by the inventory data, a combination of the current operation status of a specific type of NF included in the communication system **1** and the current operation status of the NS having the specific type of NF under the control thereof.

(285) Then, when the identified combination satisfies the condition linked to the action data, the policy manager module **140** may instruct the state machine associated with the NF to execute the action indicated by the action data on the specific type of NF. Then, the state machine may execute the action.

(286) In this case, examples of the condition indicated by the condition data linked to the action data indicating the scale-out of a specific type of NF include “the availability of a CU-CP is smaller than the predetermined value a1 and the availability of a gNB having the CU-CP under the control thereof is equal to or larger than the predetermined value a2.”

(287) Then, in this case, for example, the policy manager module **140** may identify based on the inventory data that the gNB **56** illustrated in FIG. **3** has the CU-CP **52** under the control thereof. Then, the policy manager module **140** may identify the operation status (for example, value of the availability in this case) of the CU-CP **52**. The policy manager module **140** may also identify the operation status (for example, value of the availability in this case) of the gNB **56**.

(288) Then, the policy manager module **140** may determine whether or not the value of the availability of the CU-CP **52** is smaller than the predetermined value a1. In this case, for example, it is assumed that the value of the availability of the CU-CP **52** is smaller than the predetermined value a1.

(289) In this case, the policy manager module **140** may determine whether or not the value of the availability of the gNB **56** is equal to or larger than the predetermined value a2.

(290) Then, when the value of the availability of the gNB **56** is equal to or larger than the predetermined value a2, the scale-out of the CU-CP **52** may be executed.

(291) Meanwhile, when the value of the availability of the gNB **56** is smaller than the

predetermined value a2, the scale-out of the CU-CP 52 may be inhibited from being executed.

(292) Further, for example, the action data may be linked to condition data indicating a condition relating to a combination of the operation status of a specific type of network slice and the operation status of each of a plurality of functional units (NFs) belonging to the specific type of network slice.

(293) Then, the policy manager module 140 may identify, based on the current status indicated by the inventory data, a combination of the current operation status of a specific type of network slice included in the communication system 1 and the current operation status of each of a plurality of NFs belonging to the specific type of network slice.

(294) Then, when the identified combination satisfies the condition linked to the action data, the policy manager module 140 may instruct the state machine associated with the network slice to execute the action indicated by the action data on the specific type of network slice. Then, the state machine may execute the action.

(295) In this case, examples of the condition indicated by the condition data linked to the action data indicating the scale-out of a specific type of network slice include “the availability of the network slice of an eMBB is smaller than the predetermined value a1 and the availability of all the plurality of NFs belonging to the specific type of network slice is equal to or larger than the predetermined value a2.”

(296) Then, in this case, for example, the policy manager module 140 may identify based on the inventory data that the DU 50, the CU-UP 54b, and the UPF 64b belong to the network slice 68b illustrated in FIG. 4. Then, the policy manager module 140 may identify the operation status (for example, value of the availability in this case) of the network slice 68b. The policy manager module 140 may also identify the operation statuses (for example, values of the availability in this case) of the DU 50, the CU-UP 54b, and the UPF 64b. In this case, for example, the operation statuses (for example, values of the availability in this case) of the DU 50, the CU-UP 54b, and the UPF 64b in regard to the network slice 68b may be identified.

(297) Then, the policy manager module 140 may determine whether or not the value of the availability of the network slice 68b is smaller than the predetermined value a1. In this case, for example, it is assumed that the value of the availability of the network slice 68b is smaller than the predetermined value a1.

(298) In this case, the policy manager module 140 may determine whether or not the values of the availability of the DU 50, the CU-UP 54b, and the UPF 64b are all equal to or larger than the predetermined value a2. In this case, for example, it may be determined whether or not the values of the availability of the DU 50, the CU-UP 54b, and the UPF 64b in regard to the network slice 68b are all equal to or larger than the predetermined value a2.

(299) Then, when the values of the availability of the DU 50, the CU-UP 54b, and the UPF 64b are all equal to or larger than the predetermined value a2, the scale-out of the network slice 68b may be executed. In this case, for example, when the values of the availability of the DU 50, the CU-UP 54b, and the UPF 64b in regard to the network slice 68b are all equal to or larger than the predetermined value a2, the scale-out of the network slice 68b may be executed.

(300) Meanwhile, when at least one of the values of the availability of the DU 50, the CU-UP 54b, and the UPF 64b is smaller than the predetermined value a2, the scale-out of the network slice 68b may be inhibited from being executed. In this case, for example, when at least one of the values of the availability of the DU 50, the CU-UP 54b, and the UPF 64b in regard to the network slice 68b is smaller than the predetermined value a2, the scale-out of the network slice 68b may be inhibited from being executed.

(301) Further, for example, the action data may be linked to condition data indicating a condition relating to a combination of the operation status of a specific type of functional unit (NF) and the operation status of a network slice to which the specific type of NF belongs.

(302) Then, the policy manager module 140 may identify, based on the current status indicated by

the inventory data, a combination of the current operation status of a specific type of NF included in the communication system **1** and the current operation status of the network slice to which the specific type of NF belongs.

(303) Then, when the identified combination satisfies the condition linked to the action data, the policy manager module **140** may instruct the state machine associated with the NF to execute the action indicated by the action data on the specific type of NF. Then, the state machine may execute the action.

(304) In this case, examples of the condition indicated by the condition data linked to the action data indicating the scale-out of a specific type of NF include “the availability of a UPF is smaller than the predetermined value a1 and the availability of the network slice to which the UPF belongs is equal to or larger than the predetermined value a2.”

(305) Then, in this case, for example, the policy manager module **140** may identify based on the inventory data that the UPF **64b** illustrated in FIG. **4** belongs to the network slice **68b**. Then, the policy manager module **140** may identify the operation status (for example, value of the availability in this case) of the UPF **64b**. In this case, for example, the operation status (for example, value of the availability in this case) of the UPF **64b** in regard to the network slice **68b** may be identified. The policy manager module **140** may also identify the operation status (for example, value of the availability in this case) of the network slice **68b**.

(306) Then, the policy manager module **140** may determine whether or not the value of the availability of the UPF **64b** is smaller than the predetermined value a1. In this case, for example, it may be determined whether or not the value of the availability of the UPF **64b** in regard to the network slice **68b** is smaller than the predetermined value a1. In this case, for example, it is assumed that the value of the availability of the UPF **64b** is smaller than the predetermined value a1.

(307) In this case, the policy manager module **140** may determine whether or not the value of the availability of the network slice **68b** is equal to or larger than the predetermined value a2.

(308) Then, when the value of the availability of the network slice **68b** is equal to or larger than the predetermined value a2, the scale-out of the UPF **64b** may be executed.

(309) Meanwhile, when the value of the availability of the network slice **68b** is smaller than the predetermined value a2, the scale-out of the UPF **64b** may be inhibited from being executed.

(310) In the embodiment described above, when the combination of the current operation status of a specific type of element and the current operation status of another type of element linked to the specific type of element satisfies the condition linked to the action data, the action indicated by the action data is executed. In this manner, according to this embodiment, it is possible to execute an action corresponding to an overall operation status of the communication system **1**.

(311) Further, in this embodiment, the action data may be linked to condition data indicating a condition relating to a combination of the operation status of a specific type of software element and the operation status of the hardware on which the specific type of software element is operating. As described above, the hardware on which a specific type of software element is operating is indicated by the inventory data.

(312) Then, the policy manager module **140** may identify, based on the current status indicated by the inventory data, a combination of the current operation status of a specific type of software element included in the communication system **1** and the current operation status of the hardware on which the specific type of software element is operating.

(313) Then, when the identified combination satisfies the condition linked to the action data, the E2EO module **96** may execute the action indicated by the action data on the specific type of software element. For example, when the identified combination satisfies the condition linked to the action data, the policy manager module **140** may instruct the state machine associated with the software element to execute the action indicated by the action data on the specific type of software element. Then, the state machine may execute the action.

(314) For example, the action data may be linked to condition data indicating a condition relating to a combination of the operation status of a specific type of network service (NS) and the operation status of the hardware on which the specific type of NS is operating (for example, hardware on which a container under the control of the specific type of NS is operating in this case).

(315) Then, the policy manager module **140** may identify, based on the current status indicated by the inventory data, a combination of the current operation status of a specific type of NS included in the communication system **1** and the current operation status of the hardware on which the specific type of NS is operating.

(316) Then, when the identified combination satisfies the condition linked to the action data, the policy manager module **140** may instruct the state machine associated with the NS to execute the action indicated by the action data on the specific type of NS. Then, the state machine may execute the action.

(317) In this case, examples of the condition indicated by the condition data linked to the action data indicating the scale-out of a specific type of NS include “a call success rate of an end-to-end service (E2E service) is smaller than the predetermined value a1 and an average value of a CPU usage rate of the hardware on which the specific type of NS is operating is smaller than the predetermined value a2.”

(318) Then, in this case, for example, the policy manager module **140** may identify, based on the inventory data, a server group on which a CNFC under the control of a certain E2E service is operating.

(319) Then, the policy manager module **140** may determine whether or not the value of the call success rate of the E2E service is smaller than the predetermined value a1. In this case, for example, it is assumed that the value of the call success rate is smaller than the predetermined value a1.

(320) In this case, the policy manager module **140** may identify whether or not the average value of the CPU usage rate of the server group on which the CNFC under the control of the E2E service is operating is smaller than the predetermined value a2. When the average value of the CPU usage rate is smaller than the predetermined value a2, the scale-out of the entire E2E service may be executed. Meanwhile, when the average value of the CPU usage rate is equal to or larger than the predetermined value a2, the scale-out of the entire E2E service may be inhibited from being executed.

(321) Further, for example, the action data may be linked to condition data indicating a condition relating to a combination of the operation status of a specific type of functional unit (NF) and the operation status of the hardware on which the specific type of NF is operating (for example, hardware on which a container under the control of the specific type of NF is operating in this case).

(322) Then, the policy manager module **140** may identify, based on the current status indicated by the inventory data, a combination of the current operation status of a specific type of NF included in the communication system **1** and the current operation status of the hardware on which the specific type of NF is operating.

(323) Then, when the identified combination satisfies the condition linked to the action data, the policy manager module **140** may instruct the state machine associated with the NF to execute the action indicated by the action data on the specific type of NF. Then, the state machine may execute the action.

(324) Further, for example, the action data may be linked to condition data indicating a condition relating to a combination of the operation status of a specific type of network slice and the operation status of the hardware on which the functional unit group (NF group) belonging to the specific type of network slice is operating (for example, hardware on which a container under the control of the NF group belonging to the specific type of network slice is operating in this case).

(325) Then, the policy manager module **140** may identify, based on the current status indicated by the inventory data, a combination of the current operation status of a specific type of network slice included in the communication system **1** and the current operation status of the hardware on which the NF group belonging to the specific type of network slice is operating.

(326) Then, when the identified combination satisfies the condition linked to the action data, the policy manager module **140** may instruct the state machine associated with the network slice to execute the action indicated by the action data on the specific type of network slice. Then, the state machine may execute the action.

(327) Further, for example, the action data may be linked to condition data indicating a condition relating to a combination of the operation status of a specific type of software element, the operation status of a software element under the control of the specific type of software element, and the operation status of the hardware on which the specific type of software element is operating.

(328) Then, the policy manager module **140** may identify, based on the current status indicated by the inventory data, a combination of the current operation status of a specific type of software element included in the communication system **1**, the current operation status of a software element under the control of the specific type of software element, and the current operation status of the hardware on which the specific type of software element is operating.

(329) Then, when the identified combination satisfies the condition linked to the action data, the policy manager module **140** may instruct the state machine associated with the software element to execute the action indicated by the action data on the specific type of software element. Then, the state machine may execute the action.

(330) In this case, examples of the condition indicated by the condition data linked to the action data indicating the relocation (replacement) of a specific type of NF include “the availability of an AMF is smaller than the predetermined value a1, the availability of all the CNFCs under the control of the AMF is equal to or larger than the predetermined value a2, and a notification of a predetermined type of alert has not been received from servers on which those CNFCs are operating.” It is presumed that this condition is satisfied when a failure has occurred in an NIC of one of the servers on which those CNFCs are operating. Accordingly, in general, it is desired that the relocation (replacement) of the specific type of NF be executed when this condition is satisfied.

(331) Then, in this case, for example, the policy manager module **140** may identify, based on the inventory data, a plurality of CNFCs under the control of the AMF **60** illustrated in FIG. **3**. The policy manager module **140** may also identify, based on the inventory data, servers on which those CNFCs are operating.

(332) Then, the policy manager module **140** may identify the operation status (for example, value of the availability in this case) of the AMF **60**. The policy manager module **140** may also identify the operation status (for example, value of the availability in this case) of each of the plurality of CNFCs under the control of the AMF **60**.

(333) Then, the policy manager module **140** may determine whether or not the value of the availability of the AMF **60** is smaller than the predetermined value a1. In this case, for example, it is assumed that the value of the availability of the AMF **60** is smaller than the predetermined value a1.

(334) In this case, the policy manager module **140** may determine whether or not the values of the availability of the respective plurality of CNFCs under the control of the AMF **60** are all equal to or larger than the predetermined value a2. In this case, for example, it is assumed that the values of the availability of those respective plurality of CNFCs are all equal to or larger than the predetermined value a2.

(335) In this case, the policy manager module **140** may examine whether or not a notification of the predetermined type of alert has been received from the servers on which those plurality of CNFCs are operating. In this case, when a notification of the predetermined type of alert has not been

received, the relocation (replacement) of the AMF **60** may be executed. Specifically, for example, the AMF **60** may be executed on another server, or a server on which the AMF **60** is currently operating may be separated from a cluster to which the server belongs.

(336) Meanwhile, when a notification of the predetermined type of alert has already been received, the relocation (replacement) of the AMF **60** may be inhibited from being executed.

(337) In the embodiment described above, when the combination of the operation status of a specific type of software element and the operation status of the hardware on which the specific type of software element is operating satisfies the condition linked to the action data, the action indicated by the action data is executed. In this manner, according to this embodiment, it is possible to execute an action corresponding to the overall operation status of the communication system **1**.

(338) Further, in this embodiment, even when the policy manager module **140** receives a notification of an alert indicating a predetermined abnormality from a server, in a case in which there is no software element operating on the server, any action may be inhibited from being executed.

(339) As described above, in this embodiment, the condition indicated by the condition data may be the performance index value (for example, availability). The condition indicated by the condition data may also be a condition relating to an event that occurs in an element, for example, the reception of a notification of an alert corresponding to a predetermined event from the monitoring management module **102**.

(340) That is, the condition indicated by the condition data may be a condition relating to a combination of the performance index value of a specific type of element or an event that occurs in the specific type of element, and, the performance index value of another type of element linked to the specific type of element or an event that occurs in the another type of element. The condition indicated by the condition data may also be a condition of the performance index value of a specific type of software element or an event that occurs in the specific type of software element, and, the performance index value of the hardware on which the software element is operating or an event that occurs in the hardware.

(341) Further, the specific type of element (software element) described above is not particularly limited. Examples of the specific type of element (software element) include the NS, the NF, the CNFC, the pod, the container, the network slice (NSI), and the network slice subnet instance (NSSI).

(342) [Execution of Action Involving Identification of Cause of Event]

(343) Now, examples of identification of a cause of a detected event and execution of an action corresponding to the identification of the cause, which are performed by the E2EO module **96**, are further described.

(344) In this embodiment, when a failure or another event occurs in a certain element included in the communication system **1**, a problem sometimes fails to be solved irrespective of countermeasures for the element, which includes healing.

(345) For example, when a root cause of an event that has occurred in the RAN **38** resides in the core network system **40** that includes this RAN **38** in an influence range of the core network system **40**, the problem is not solved even by performing the healing on this RAN **38**, and there is even a fear in that a multi-failure may rather be caused.

(346) Accordingly, in order to execute appropriate healing, it is required to accurately identify the cause of an event that has occurred in the communication system **1**.

(347) In consideration of this, in this embodiment, in such a manner as described below, the cause of an event that has occurred in the communication system **1** may be enabled to be accurately identified.

(348) In this embodiment, the inventory data may indicate the current status of the influence relationship between the elements included in the communication system **1** (for example, geographical relationship or topological relationship between the elements).

(349) As illustrated in FIG. 11 to FIG. 14, it can be said that the above-mentioned inventory data indicates the current status of the topological relationship between the elements.

(350) The above-mentioned inventory data also includes location data indicating locations at which the elements included in the communication system 1 are operating, that is, the current locations of the elements included in the communication system 1. It can be said therefrom that the above-mentioned inventory data indicates the current status of the geographical relationship between the elements (for example, geographical closeness between the elements).

(351) Further, as described above, the policy manager module 140 in this embodiment stores the action data indicating actions to be executed on the elements included in the communication system 1.

(352) It is also assumed that the action data described below is linked to event type data indicating a type of an event.

(353) The policy manager module 140 in this embodiment may identify, based on the inventory data, the influence relationship (for example, geographical closeness or topological closeness) between the elements included in the communication system 1.

(354) The policy manager module 140 in this embodiment may also detect the occurrence of an event in a specific element included in the communication system 1. For example, in response to a confirmation that the performance index value relating to the specific element is larger than a predetermined threshold value or the performance index value relating to the specific element is smaller than a predetermined threshold value, which has been received from the monitoring management module 102, the policy manager module 140 may detect the occurrence of the event corresponding to this fact in the specific element. In another case, the policy manager module 140 may detect, in response to the reception of the notification of an alert relating to a certain element, the occurrence of an event corresponding to the alert in the element. The element for which the occurrence of an event has been detected in this manner is hereinafter referred to as “event occurrence element.”

(355) Then, the policy manager module 140 may identify, based on the inventory data, at least one other element having an influence relationship with the event occurrence element. The other element having an influence relationship with the event occurrence element identified in this manner is hereinafter referred to as “influence element.”

(356) In this case, for example, the policy manager module 140 may identify at least one influence element relating to the event occurrence element based on the geographical closeness or the topological closeness between each of the elements included in the communication system 1 and the event occurrence element, which is indicated by the inventory data.

(357) Specifically, for example, a server or data center at a location away from the location of a server or data center on which the event occurrence element is operating by a predetermined distance or shorter may be identified. Then, the element operating on the identified server or data center may be identified as the influence element relating to the event occurrence element.

(358) In another case, an element belonging to the same network slice as that of the event occurrence element may be identified as the influence element relating to the event occurrence element. In another case, a parent element having the event occurrence element under the control thereof or a plurality of elements under the control of the parent element may be identified as the influence elements relating to the event occurrence element.

(359) For example, it is assumed that the occurrence of an event in the RAN 38 has been detected. In this case, the policy manager module 140 may identify, as the influence element relating to the RAN 38, the core network system 40 having an influence relationship with the RAN 38 being the event occurrence element.

(360) In this case, for example, the policy manager module 140 may identify, as the influence element relating to the RAN 38, the core network system 40 belonging to the same network slice as that of the RAN 38.

(361) Further, for example, the policy manager module **140** may identify, as the influence element relating to the RAN **38**, the core network system **40** geographically or topologically closest to the RAN **38** from among a plurality of core network systems **40** included in the communication system **1**. For example, the core network system **40** present at the location closest to the RAN **38** may be identified as the influence element relating to the RAN **38**. Further, for example, the core network system **40** having the smallest number of nodes between the core network system **40** and the RAN **38** indicated by the topology data may be identified as the influence element relating to the RAN **38**.

(362) Then, the policy manager module **140** may identify the cause of the event (for example, element being the cause of the event) based on a status of at least one influence element relating to the event occurrence element. The element identified as the cause of the event is hereinafter referred to as “causative element.”

(363) FIG. **16** and FIG. **17** are diagrams for schematically illustrating an example of an event occurrence situation. For example, it is assumed in both FIG. **16** and FIG. **17** that the RAN **38(1)** has been detected as the event occurrence element (for example, element in which a failure occurrence event has occurred in this case).

(364) Then, as illustrated in FIG. **16** and FIG. **17**, it is assumed that the core network system **40(1)** and the RAN **38(2)** to the RAN **38(n)** are identified as the influence elements for RAN **38(1)**.

(365) In this case, the policy manager module **140** identifies the status of each of the influence elements. In this case, for example, it is identified whether or not a failure has occurred.

(366) Then, the policy manager module **140** identifies any one of the RAN **38(1)** or the core network system **40(1)** as the causative element based on whether or not those influence elements satisfy a predetermined condition.

(367) For example, as illustrated in FIG. **16**, it is assumed that it has been confirmed that a failure has occurred in all the influence elements (for example, core network system **40(1)** and RAN **38(2)** to RAN **38(n)** in this case). In this case, the core network system **40(1)** is identified as the causative element.

(368) Otherwise, the RAN **38(1)** is identified as the causative element. For example, as illustrated in FIG. **17**, when it is confirmed that a failure has occurred only in the RAN **38(4)** among the influence elements, the RAN **38(1)** is identified as the causative element. In this case, the RAN **38(1)** and the RAN **38(4)** may be identified as the causative elements.

(369) For example, as described above, the policy manager module **140** may identify whether the cause of the event resides in the core network system **40** or in the RAN **38** based on the statuses of the core network system **40** and all the RANs **38** in the influence range of the core network system **40**.

(370) In this case, the policy manager module **140** may identify the cause of the event based on the status of the core network system **40**. For example, the policy manager module **140** may identify whether the cause of the event resides in the core network system **40** or in the RAN **38** based on the status of the core network system **40**. For example, when it is confirmed that a failure has occurred in the core network system **40**, the core network system **40** may be identified as the causative element. Meanwhile, when it is confirmed that no failure has occurred in the core network system **40**, the RAN **38** may be identified as the causative element.

(371) Further, in this embodiment, the policy manager module **140** may transmit a predetermined packet for confirmation (for example, ping) to each of at least one influence element. Then, the policy manager module **140** may identify the cause of the event based on whether or not the packet successfully reaches each of at least one influence element. For example, an element that has not been reached by the packet for confirmation may be identified as the causative element.

(372) Further, for example, the packet for confirmation may be transmitted to the influence element in which the occurrence of a failure has been confirmed in such a manner as described above.

Then, of the influence elements in which the occurrence of a failure has been confirmed, an

influence element that has not been reached by the packet for confirmation may be identified as the causative element.

(373) Then, the E2EO module **96** may execute an action such as the healing corresponding to the event on the causative element identified as described above. In this case, for example, the policy manager module **140** may identify the action data linked to the event type data indicating the type of the event. Then, the policy manager module **140** may output the execution instruction for the action indicated by the identified action data to the state machine corresponding to the identified element. Then, the state machine may start to execute the workflow associated with the action on the element.

(374) For example, in the status illustrated in FIG. **16**, even when the healing is executed on the RAN **38(1)**, the problem is not solved as long as the cause of the event resides in the core network system **40(1)**. Even in such a case, according to this embodiment, the cause of an event that has occurred in the communication system **1** is accurately identified in such a manner as described above, and hence the problem involved in the occurrence of the event is appropriately solved.

(375) [Determination of Location]

(376) In this embodiment, when the scale-out or the replacement is executed, a new element is added to the communication system **1**. Now, an example of determination of a location at which a new element added in this manner is to be constructed is described.

(377) In the following description, it is assumed that the communication system **1** includes a core network system **150** illustrated in FIG. **18**.

(378) There are a plurality of NSes (for example, AMF service **152**, SMF service **154**, and UPF service **156**) under the control of the core network system **150** illustrated in FIG. **18**. As described above, the core network system **150** corresponds to an element having a granularity of an NS, and hence there are NSes under the control of the NS in this example. In this manner, in this embodiment, NSes may be provided in a nested structure.

(379) Further, as illustrated in FIG. **18**, the AMF service **152** includes a plurality of AMFs **158** (**158a**, **158b**, . . .). In addition, the SMF service **154** includes a plurality of SMFs **160** (**160a**, **160b**, . . .). Further, the UPF service **156** includes a plurality of UPFs **162** (**162a**, **162b**, . . .).

(380) FIG. **19** is a diagram for illustrating an example of the topology data for a plurality of elements, which indicates the current statuses of links between the elements included in the core network system **150** illustrated in FIG. **18**.

(381) In FIG. **19**, the topology data is represented as data including nodes **170** associated with elements included in the communication system **1** and links **172** each connecting the nodes **170** associated with respective two elements linked to each other. More specifically, in FIG. **19**, the topology data is represented as tree structure data in which a hierarchical structure between the elements is represented by a parent-child relationship.

(382) A node **170a** having an identifier of “Core #201,” which is illustrated in FIG. **19**, is associated with the core network system **150** illustrated in FIG. **18**. A node **170b** having an identifier of “AMF service #201” is associated with the AMF service **152** illustrated in FIG. **18**. A node **170c** having an identifier of “SMF service #201” is associated with the SMF service **154** illustrated in FIG. **18**. A node **170d** having the identifier of “UPF service #201” is associated with the UPF service **156** illustrated in FIG. **18**.

(383) A node **170e** having an identifier of “AMF #201” is associated with the AMF **158a** illustrated in FIG. **18**. A node **170f** having an identifier of “AMF #202” is associated with the AMF **158b** illustrated in FIG. **18**.

(384) A node **170g** having an identifier of “SMF #201” is associated with the SMF **160a** illustrated in FIG. **18**. A node **170h** having an identifier of “SMF #202” is associated with the SMF **160b** illustrated in FIG. **18**.

(385) A node **170i** having an identifier of “UPF #201” is associated with the UPF **162a** illustrated in FIG. **18**. A node **170j** having an identifier of “UPF #202” is associated with the UPF **162b**

illustrated in FIG. 18.

(386) FIG. 19 shows a link 172a connecting the node 170a and the node 170b that is a node 170 being a child of the node 170a. FIG. 19 also shows a link 172b connecting the node 170a and the node 170c that is a node 170 being a child of the node 170a. FIG. 19 also shows a link 172c connecting the node 170a and the node 170d that is a node 170 being a child of the node 170a.

(387) FIG. 19 also shows a link 172d connecting the node 170b and the node 170e that is a node 170 being a child of the node 170b. FIG. 19 also shows a link 172e connecting the node 170b and the node 170f that is a node 170 being a child of the node 170b.

(388) FIG. 19 also shows a link 172f connecting the node 170c and the node 170g that is a node 170 being a child of the node 170c. FIG. 19 also shows a link 172g connecting the node 170c and the node 170h that is a node 170 being a child of the node 170c.

(389) FIG. 19 also shows a link 172h connecting the node 170d and the node 170i that is a node 170 being a child of the node 170d. FIG. 19 also shows a link 172i connecting the node 170d and the node 170j that is a node 170 being a child of the node 170d.

(390) In this embodiment, for example, it is assumed that the policy manager module 140 determines that a new element linked to an already installed element included in the communication system 1 is to be added to the communication system 1. For example, as described above, it is assumed that the policy manager module 140 determines that the scale-out or the replacement is to be executed based on the performance index value or the alert received from the monitoring management module 102.

(391) In this case, in response to the determination, the resource manager module 142 may determine a location at which the new element is to be constructed based on new topology data including the new element and a current location of the already installed element indicated by the location data. In this case, the resource manager module 142 may determine a location of a cluster in which a new element is to be constructed.

(392) Now, an exemplary case in which it has been determined that a new AMF 158 being a new functional unit is to be added under the control of the core network system 150 being an already installed network service is described.

(393) In this case, the topology data illustrated in FIG. 19 is updated to the topology data illustrated in FIG. 20. In the topology data illustrated in FIG. 20, a node 170k having an identifier of “AMF #301” has been added. In addition, a link 172j connecting the node 170b and the node 170k has been added. A planned attribute is set for the node 170k and the link 172j.

(394) Then, the resource manager module 142 may first determine whether or not the new AMF 158 can be constructed in a cluster (for example, Kubernetes cluster) in which an already installed AMF 158 is constructed.

(395) For example, the resource manager module 142 may identify a current availability status of the cluster in which the already installed AMF 158 is constructed.

(396) Then, the resource manager module 142 may identify computer resources required for constructing the AMF 158 based on the CNFD or other service template data. As described above, the CNFD or other service template data indicates, for each of a plurality of containers included in the AMF 158, the computer resources required by the container. Accordingly, the resource manager module 142 can identify the computer resources required for constructing the AMF 158 based on, for example, the CNFD or other service template data.

(397) Then, the resource manager module 142 may determine whether or not the new AMF 158 can be constructed in the cluster based on the computer resources required for constructing the new AMF 158 and the availability status of the cluster in which the already installed AMF 158 is constructed. For example, when the computer resources required for constructing the new AMF 158 can be secured in the cluster in which the already installed AMF 158 is constructed, the resource manager module 142 may determine that the new AMF 158 can be constructed in the cluster.

(398) In this case, when it is determined that the new AMF **158** can be constructed, the resource manager module **142** determines a location of the cluster as a location of the new AMF **158**.

(399) In contrast, when it is determined that the new AMF **158** cannot be constructed, the resource manager module **142** may identify a specific type of element linked to the AMF **158** based on new topology data including the node **170k** and the link **172j**.

(400) In this case, for example, the above-mentioned NSD or other service template data may indicate a performance requirement relating to communication between a specific plurality of types of elements. For example, the performance requirement that “latency (communication time period) between an AMF and an SMF included in a core network system is equal to or shorter than 5 milliseconds” may be indicated by the NSD associated with the core network system.

(401) Then, the resource manager module **142** may identify a type of element to be identified based on the NSD associated with the core network system. In this case, for example, the NSD indicates the performance requirement relating to the communication between the AMF and the SMF, and hence the SMF is identified as the type of element to be identified.

(402) Then, the resource manager module **142** may identify the SMF linked to the AMF **158** associated with the node **170k** based on the topology data.

(403) In the topology data illustrated in FIG. **20**, the functional units under the control of a common network service are linked to each other. For example, the AMF **158a**, the AMF **158b**, the new AMF **158**, the SMF **160a**, and the SMF **160b** under the control of the common core network system **150** are linked to each other.

(404) In this case, the resource manager module **142** may identify the SMF **160a** and the SMF **160b** as the SMFs linked to the new AMF **158** based on the topology data.

(405) Then, the resource manager module **142** may identify current locations of the already installed functional units (for example, SMF **160a** and SMF **160b** in this case) identified in this manner, which are indicated by the above-mentioned location data.

(406) Then, the resource manager module **142** may determine a location at which a new element (for example, new AMF **158** in this case) is to be constructed based on the current locations of the already installed functional units identified in this manner.

(407) For example, the resource manager module **142** may identify the location of the cluster at which the SMF **160a** and the SMF **160b** are constructed. Then, the resource manager module **142** may use, for example, a known geographic information service to identify clusters constructed at such locations that a distance from the location identified in this manner is equal to or smaller than a predetermined distance. Then, a cluster in which the new AMF **158** can be constructed may be identified from among the clusters identified in this manner. Then, the location of the cluster identified in this manner may be determined as the location in which the new AMF **158** is to be constructed.

(408) In this case, the resource manager module **142** may determine a location satisfying a given performance requirement relating to communication between the already installed element and the new element as the location at which the new element is to be constructed.

(409) For example, it is assumed that, as described above, the NSD associated with a core network system indicates the performance requirement that “latency (communication time period) between an AMF and an SMF included in a core network system is equal to or shorter than 5 milliseconds.” In this case, for example, a method similar to cost calculation for routing may be used to identify such clusters that the time period of communication to/from the cluster in which the SMF **160a** and the SMF **160b** are constructed is equal to or shorter than 5 milliseconds. Then, the cluster in which the new AMF **158** can be constructed may be identified from among the clusters identified in this manner. Then, the location of the cluster identified in this manner may be determined as the location at which the new AMF **158** is to be constructed. The given performance requirement is not limited to the latency. For example, the given performance requirement may be a performance requirement relating to throughput in communication between the AMF and the SMF. For example,

a condition (for example, minimum value) to be satisfied by the throughput may be indicated by the NSD associated with the core network system.

(410) In the above-mentioned example, the resource manager module **142** may determine a location different from a location of a cluster in which already installed elements linked to a new element in the topology data are constructed as the location of the cluster in which the new element is to be constructed. For example, the resource manager module **142** may determine a location different from the location of the cluster in which the SMF **160a** and the SMF **160b** are constructed as the location at which the new AMF **158** is to be constructed.

(411) Further, the resource manager module **142** may determine the location of the cluster in which the SMF **160a** and the SMF **160b** are constructed as the location at which the new AMF **158** is to be constructed.

(412) Further, a label associated with an element that can be constructed may be set for each of clusters included in the communication system **1**. Then, the resource manager module **142** may determine the cluster in which the new element is to be constructed from among clusters for which the label associated with the new element is set.

(413) As described above, the policy manager module **140** outputs an execution instruction for an action to the state machine corresponding to an element on which the execution of the action has been determined. Then, the state machine starts to execute the workflow associated with the action on the element.

(414) In this case, the policy manager module **140** may output, to the state machine, an instruction to construct a new element at the location determined by the resource manager module **142**. Then, the state machine may start to execute the workflow associated with the construction of the new element at the location.

(415) Then, the E2EO module **96**, the configuration management module **100**, and the container management module **106** may construct the new element at the location determined as described above.

(416) As described above, for example, the new AMF **158** is added under the control of the AMF service **152**. Then, the setting of the planned attribute regarding the node **170k** and the link **172j** included in the topology data illustrated in FIG. **20** is canceled.

(417) In this manner, in this embodiment, for example, the scale-out of the AMF **158** is executed, to thereby update the inventory data including the location data and the topology data.

(418) In this embodiment, the resource manager module **142** is not required to determine the location of the cluster in which a new element is to be constructed.

(419) For example, the resource manager module **142** may determine the location of a server on which a new element is to be constructed. In this case, the resource manager module **142** may determine a location different from a location of a server on which already installed elements linked to a new element in the topology data are constructed as the location of the server on which the new element is to be constructed. Then, the new element may be constructed on the server at the location determined in this manner.

(420) Further, for example, the resource manager module **142** may determine the location of a data center in which a new element is to be constructed. In this case, the resource manager module **142** may determine a location different from a location of a data center in which already installed elements linked to a new element in the topology data are constructed as the location of the data center in which the new element is to be constructed. Then, the new element may be constructed in the data center at the location determined in this manner.

(421) Further, as illustrated in FIG. **12** and FIG. **14**, in the topology data, functional units belonging to a common network slice may be linked to each other.

(422) In this case, it is assumed that it is determined that a new functional unit is to be added to the communication system **1** as the functional unit belonging to an already installed network slice included in the communication system **1**. For example, it is assumed that it is determined that the

UPF **64** belonging to the network slice **68b** illustrated in FIG. **4** is to be added.

(423) In this case, the resource manager module **142** may identify the already installed functional unit indicated in the topology data as the functional unit belonging to the network slice. In this case, for example, the CU-UP **54b** belonging to the network slice **68b** may be identified.

(424) Then, the resource manager module **142** may identify a current location of the identified already installed functional unit, which is indicated by the location data. In this case, for example, the current location of the CU-UP **54b** may be identified.

(425) Then, the resource manager module **142** may determine a location at which a new functional unit is to be constructed based on the current locations of the already installed functional units identified in this manner. In this case, for example, clusters at such locations that a distance from the location of the cluster in which the CU-UP **54b** is constructed is equal to or smaller than a predetermined distance may be identified. In another case, for example, such clusters that the time period of communication to/from the cluster in which the CU-UP **54b** is constructed is equal to or shorter than a predetermined time period may be identified. Then, a cluster in which the new UPF **64** can be constructed may be identified from among the clusters identified in this manner. Then, the location of the cluster identified in this manner may be determined as the location in which the new UPF **64** is to be constructed.

(426) When the location of an element to be newly constructed in the communication system **1** is determined at random or in a round robin, there is a fear in that the element may be constructed at an undesired location. For example, there is a fear in that the AMF **158** to be newly added under the control of a certain core network system **150** may be constructed at a location far from an already installed SMF **160** under the control of the core network system **150**. Then, as a result, there is a fear in that there may occur performance degradation including deterioration of latency and reduction of throughput.

(427) In this embodiment, as described above, the location of a new element is determined based on the current location of an already installed element linked to the new element. Accordingly, for example, the new AMF **158** to be added to the core network system **150** is to be constructed at a location near the already installed SMF **160** included in the core network system **150**.

(428) As described above, according to this embodiment, it is possible to appropriately determine a location at which a new element is to be constructed in the communication system **1**.

(429) The above-mentioned element for which the location is to be determined is not particularly limited. Examples of the element for which the location is to be determined include the NS, the NF, the CNFC, the pod, the container, the network slice (NSI), and the network slice subnet instance (NSSI).

(430) [Flows of Processes]

(431) An example of flows of processes executed by the vendor terminal **36**, the MPS **30**, and the NOS **32** when the “ONBOARDING” button **70** is clicked by a vendor on the onboarding screen illustrated in FIG. **7** is described with reference to flow charts illustrated in FIG. **21A** and FIG. **21B**.

(432) First, the vendor terminal **36** transmits bundle data arranged in a path specified on the onboarding screen to the bundle development module **90** of the NOS **32** (Step **S101**).

(433) Then, the bundle development module **90** develops the bundle data received in the process step of Step **S101** and generates a data group illustrated in FIG. **9** (Step **S102**).

(434) Then, the bundle development module **90** determines a bundle ID corresponding to the data group generated in the process step of Step **S102** (Step **S103**).

(435) The bundle development module **90** then transmits product catalog data included in the data group generated in the process step of Step **S102** that is linked to the bundle ID determined in the process step of Step **S103** to the product management module **80** of the MPS **30**. Then, the product management module **80** of the MPS **30** stores the received product catalog data in the product catalog storage **82** (Step **S104**).

(436) The bundle development module **90** then outputs to the service management module **92**

service catalog data included in the data group generated in the process step of Step S102 that is linked to the bundle ID determined in the process step of Step S103. Then, the service management module **92** stores the received service catalog data in the service catalog storage **94** (Step S105).

(437) The bundle development module **90** stores inventory template data included in the data group generated in the process step of Step S102 that is linked to the bundle ID determined in the process step of Step S103 in the inventory management module **98** (Step S106).

(438) The bundle development module **90** then stores CM template data included in the data group illustrated in Step S102 that is linked to the bundle ID determined in the process step of Step S103 (Step S107).

(439) The bundle development module **90** stores monitoring script data included in the data group generated in the process step of Step S102 that is linked to the bundle ID determined in the process step of Step S103 in the monitoring management module **102** (Step S108).

(440) Then, the bundle development module **90** stores security script data included in the data group generated in the process step of Step S102 that is linked to the bundle ID determined in the process step of Step S103 in the security setting module **104** (Step S109).

(441) The bundle development module **90** then stores Helm chart data included in the data group generated in the process step of Step S102 that is linked to the bundle ID determined in the process step of Step S103 (Step S110) in the container management module **106**. For example, the bundle development module **90** may store the Helm chart included in the data group generated in the process step of Step S102 in a plurality of container management modules **106**. Further, Helm chart data associated with the container management module **106** may be stored in the container management module **106**.

(442) Then, the bundle development module **90** stores container image data included in the data group generated in the process step of S102 that is linked to the bundle ID determined in the process step of Step S103 (Step S111) in the repository module **108**, and the process illustrated in this process example is ended.

(443) Next, an example of a flow of processes executed by the purchaser terminal **34**, the MPS **30**, and the NOS **32** in accordance with a purchase operation of a network service, which is performed by the purchaser, is described with reference to a flow chart illustrated in FIG. 22.

(444) First, the purchaser terminal **34** transmits a purchase request for a network service linked to the purchase bundle ID and the purchase service requirement data to the purchase management module **84** of the MPS **30** (Step S201).

(445) Then, the purchase management module **84** transmits to the E2EO module **96** the purchase request linked to the purchase bundle ID and the purchase service requirement data received in the process step of Step S201 (Step S202).

(446) The E2EO module **96** then identifies service catalog data corresponding to the purchase bundle ID linked to the received purchase request (Step S203).

(447) Then, the E2EO module **96** acquires the service catalog data identified in the process step of Step S203 from the service catalog storage **94** and executes the workflow script indicated by the service catalog data (Step S204), and the process illustrated in this process example is ended.

(448) The details of the process step of Step S204 is now described with reference to flow charts illustrated in FIG. 23A to FIG. 23F.

(449) First, the E2EO module **96** and the inventory management module **98** generate planned data based on the purchase service requirement data linked to the purchase request, service catalog data, inventory template data, and inventory data (Step S301).

(450) Then, the inventory management module **98** stores the generated planned data in the active inventory **114** (Step S302).

(451) Then, the inventory management module **98** stores the generated planned data in the inventory database **110** (Step S303).

(452) The inventory management module **98** then outputs an inventory key included in the

generated planned data to the E2EO module **96** (Step S304).

(453) Then, the E2EO module **96** outputs the inventory key that has been received to the configuration management module **100** (Step S305).

(454) The configuration management module **100** then acquires planned data including the received inventory key from the active inventory **114** (Step S306). In this case, the planned data may be acquired from the inventory database **110**.

(455) Then, the configuration management module **100** generates and holds planned CM data including a day 1 parameter based on the planned data acquired in the process step of Step S306 (Step S307).

(456) The configuration management module **100** outputs an instruction for a setup, for example, securing required hardware resources, to the bare metal management module **112** (Step S308), and the bare metal management module **112** executes a setup, for example, securing hardware resources in accordance with the instruction (Step S309). At this time, the setup of system software in accordance with a specific type of functional unit and addition of a free server to the resource pool are executed as required.

(457) In this embodiment, a free server may be added to the resource pool with an enough margin (buffer). For example, a plurality of servers **120** may be collectively added to the resource pool.

(458) Then, the bare metal management module **112** outputs a completion notification to the configuration management module **100** (Step S310).

(459) Then, the configuration management module **100** outputs a completion notification to the E2EO module **96** (Step S311).

(460) Then, the E2EO module **96** acquires the planned data including the inventory key received in the process step of Step S304 from the active inventory **114** (Step S312). In this case, the planned data may be acquired from the inventory database **110**.

(461) Then, the E2EO module **96** identifies a location at which the element group is deployed based on the planned data acquired in the process step of Step S312 (Step S313).

(462) The E2EO module **96** then generates a day 0 parameter (CNF instance) for each location identified in the process step of Step S313 (Step S314).

(463) Then, to a container management module **106** corresponding to each location identified in the process step of Step S313, the E2EO module **96** outputs a day 0 parameter corresponding to the container management module **106** (Step S315).

(464) The container management module **106** then executes the deployment of a container based on the day 0 parameter that has been received (Step S316).

(465) Then, the container management module **106** outputs the completion notification to the E2EO module **96** (Step S317).

(466) The E2EO module **96** then outputs a configuration management instruction that is based on the day 1 parameter to the configuration management module **100** (Step S318).

(467) Then, the configuration management module **100** executes the configuration management of a container group that is based on the day 1 parameter included in the held planned CM data (Step S319).

(468) The configuration management module **100** then outputs a completion notification to the E2EO module **96** (Step S320).

(469) Then, the E2EO module **96** executes the instantiation of a network slice based on the planned data acquired in the process step of Step S312 (Step S321). In the process step of Step S321, for example, as described above, the E2EO module **96** may output a configuration management instruction related to the instantiation of the network slice to the configuration management module **100**. Then, the configuration management module **100** may execute configuration management, for example, settings in accordance with the configuration management instruction.

(470) Further, as described above, the configuration management module **100** may update the day 1 parameter based on the configuration management instruction received from the E2EO module **96**

in the process step of Step S321 without executing the process steps of from Step S318 to Step S320. Then, the configuration management module **100** may execute configuration management, for example, settings in accordance with the configuration management instruction.

(471) The E2EO module **96** then outputs the inventory key received in the process step of Step S304 to the monitoring management module **102** (Step S322).

(472) Then, the monitoring management module **102** acquires planned data including the received inventory key from the active inventory **114** (Step S323). In this case, the planned data may be acquired from the inventory database **110**.

(473) Based on the planned data acquired in the process step of Step S323, the monitoring management module **102** then executes a monitoring setting in accordance with a monitoring policy indicated by the purchase service requirement data (Step S324).

(474) Then, the monitoring management module **102** outputs a completion notification to the E2EO module **96** (Step S325).

(475) The E2EO module **96** then outputs the inventory key received in the process step of Step S304 to the security setting module **104** (Step S326).

(476) Then, the security setting module **104** acquires planned data including the received inventory key from the active inventory **114** (Step S327). In this case, the planned data may be acquired from the inventory database **110**.

(477) The security setting module **104** then executes a security setting based on the planned data acquired in the process step of Step S327 (Step S328).

(478) Then, the security setting module **104** outputs a completion notification to the E2EO module **96** (Step S329).

(479) Then, the E2EO module **96** outputs a completion notification to the inventory management module **98** (Step S330).

(480) Then, the inventory management module **98** updates the inventory data so that the setting of the planned attribute regarding the planned data stored in the active inventory **114** and the inventory database **110** is canceled (Step S331), and the process illustrated in this process example is ended.

(481) Next, an example of a flow of a process executed by the monitoring management module **102** of the NOS **32** is described with reference to a flow chart illustrated in FIG. 24. FIG. 24 is an illustration of a process for calculating a desired performance index value for one specific element based on one specific piece of calculation logic data.

(482) First, the monitoring management module **102** identifies the condition to be satisfied by an element used for calculating the desired performance index value in the calculation logic data (Step S401).

(483) Then, the monitoring management module **102** searches the inventory data to identify one or a plurality of elements that satisfy the condition identified in the process step of Step S401 (Step S402). In the process step of Step S402, for example, an element corresponding to the node **130** being a child of the node **130** corresponding to the element for which the desired performance index value is to be calculated is identified.

(484) Then, the monitoring management module **102** substitutes the element identified in the process step of Step S402 as the value of the parameter of the calculation logic indicated by the calculation logic data (Step S403).

(485) Then, the monitoring management module **102** identifies the performance index value indicated as the monitoring item for the element in the calculation logic data for the element substituted as the value of the parameter in the process step of Step S403 (Step S404).

(486) Then, the monitoring management module **102** calculates the desired performance index value by applying the calculation logic indicated by the calculation logic data to the performance index value identified in the process step of Step S404 (Step S405), and the process illustrated in this process example is ended.

(487) Next, an example of a flow of an execution process of an action executed by the E2EO module **96** of the NOS **32** is described with reference to a flow chart illustrated in FIG. 25. FIG. 25 is an illustration of a process for executing an action indicated by the action data based on one specific piece of action data.

(488) First, the policy manager module **140** identifies the operation status indicated by the condition data linked to the action data (Step S501). In this case, the policy manager module **140** identifies the operation status indicated by the condition data linked to the action data based on data indicating the operation status, for example, data indicating the performance index value or data indicating the notification of the alert, which is held by the policy manager module **140**.

(489) As described above, in the process step of Step S501, a combination of the operation status of a specific type of element and the data indicating the operation status of another type of element linked to the specific type of element may be identified. Further, the combination of the operation status of a specific type of software element and the operation status of the hardware on which the specific type of software element is operating may be identified.

(490) Then, the policy manager module **140** determines whether or not the operation status identified in the process step of Step S501 satisfies the condition linked to the action data (Step S502).

(491) When it is determined that the condition is not satisfied (“N” in Step S502), the process illustrated in this process example is ended.

(492) When it is determined that the condition is satisfied (“Y” in Step S502), the E2EO module **96** executes the action indicated by the action data (Step S503), and the process illustrated in this process example is ended.

(493) In the process step of Step S503, for example, the policy manager module **140** may instruct the state machine associated with the specific type of element indicated by the action data to execute the action indicated by the action data. Then, the state machine may execute the action.

(494) The process steps of from Step S501 to Step S503 illustrated in FIG. 25 are executed on, for example, all pieces of action data stored in the policy manager module **140** at a predetermined timing. For example, the process steps of from Step S501 to Step S503 may be executed at a predetermined time interval, or may be executed in response to the reception of the data indicating the operation status, for example, the reception of the data indicating the performance index value or the notification of the alert.

(495) Next, an example of a flow of a process for identifying the cause of an event and a process for executing the action, which are executed by the E2EO module **96** of the NOS **32** in response to the detection of the occurrence of the event relating to the event occurrence element, is described with reference to a flow chart illustrated in FIG. 26.

(496) First, the policy manager module **140** monitors the occurrence of an event for each of the elements included in the communication system **1** (Step S601).

(497) When it is detected that an event has occurred in a certain element (event occurrence element), the policy manager module **140** identifies at least one influence element relating to the event occurrence element based on the inventory data (Step S602).

(498) Then, the policy manager module **140** examines the status of the influence element identified in the process step of Step S602 (Step S603).

(499) Then, the policy manager module **140** identifies the causative element based on the status of the influence element examined in the process step of Step S603 (Step S604).

(500) Then, the E2EO module **96** executes the action corresponding to the event on the causative element identified in the process step of Step S604 (Step S605), the process returns to the process step of Step S601.

(501) In the process step of Step S605, for example, the policy manager module **140** may instruct the state machine associated with the causative element to execute the action corresponding to the event. Then, the state machine may execute the action.

(502) As a part of the examination of the status indicated in Step S603, the packet for confirmation may be transmitted to the influence element.

(503) In another case, the packet for confirmation may be transmitted to the influence element in which the occurrence of a failure has been confirmed between the process step of Step S603 and the process step of Step S604. Then, in the process step of Step S604, of the influence elements in which the occurrence of a failure has been confirmed, the element that has not been reached by the packet for confirmation may be identified as the causative element.

(504) Next, an example of a flow of a process for the determination of the location, which is executed by the resource manager module 142 in response to the determination that a new element linked to the already installed element included in the communication system 1 is to be added to the communication system 1, is described with reference to a flow chart illustrated in FIG. 27. FIG. 27 is an illustration of a process executed in response to the determination that the new AMF 158 linked to the already installed SMF 160 included in the core network system 150 illustrated in FIG. 18 is to be added to the core network system 150.

(505) First, the inventory management module 98 updates the topology data illustrated in FIG. 19 to the topology data illustrated in FIG. 20 (Step S701). In this case, the planned attribute is set for the node 170k and the link 172j.

(506) Then, the resource manager module 142 identifies computer resources required for constructing the new AMF 158 based on the CNFD (Step S702).

(507) Then, the resource manager module 142 identifies the availability status of the cluster in which the already installed AMFs 158 (AMF 158a and AMF 158b) are constructed (Step S703).

(508) Then, the resource manager module 142 determines whether or not the new AMF 158 can be constructed in the cluster in which the already installed AMFs 158 are constructed based on the computer resources required for constructing the new AMF 158, which are identified in the process step of Step S702, and the availability status identified in the process step of Step S703 (Step S704).

(509) It is assumed that it has been determined that the new AMF 158 can be constructed in the cluster in which the already installed AMFs 158 are constructed (Y in Step S704). In this case, the resource manager module 142 determines the location of the cluster as the location of the new AMF 158 (Step S705), and the process advances to Step S711.

(510) It is assumed that it has been determined that the new AMF 158 cannot be constructed in the cluster in which the already installed AMFs 158 are constructed (N in Step S704). In this case, the resource manager module 142 identifies an element linked to the new AMF 158 based on the NSD associated with the core network system and the topology data (Step S706). In the process step of Step S706, for example, the SMF 160a and the SMF 160b are identified as the elements linked to the new AMF 158.

(511) Then, the resource manager module 142 identifies, based on the location data, the current location of the element identified in the process step of Step S706, which is indicated by the location data (Step S707). In the process step of Step S707, for example, the current location of the cluster in which the SMF 160a and the SMF 160b are constructed is identified.

(512) Then, the resource manager module 142 identifies candidates for the cluster in which the new AMF 158 is to be constructed based on the location identified in the process step of Step S707 (Step S708).

In the process step of Step S708, for example, clusters at such locations that a distance from the location of the cluster in which the SMF 160a and the SMF 160b are constructed is equal to or smaller than a predetermined distance may be identified. In addition, for example, clusters at locations satisfying the performance requirement, which is indicated by the NSD associated with the core network system, with the location identified in the process step of Step S707 being used as a reference may be identified.

(513) Then, the resource manager module 142 identifies the cluster in which the new AMF 158 can be constructed from among the candidates for the cluster identified in the process step of Step S708

(Step S709). In the process step of Step S709, for example, the cluster in which the new AMF 158 can be constructed may be identified based on the computer resources required for constructing the new AMF 158, which are identified in the process step of Step S702, and the availability status of each of the candidates for the cluster.

(514) Then, the resource manager module 142 determines the location of the cluster identified in the process step of Step S709 as the location of the new AMF 158 (Step S710). In the process step of Step S709, there is a possibility that a plurality of clusters may be identified. In this case, the location of any one of the clusters determined in accordance with a predetermined rule from among those clusters may be determined as the location of the new AMF 158. For example, the location of any one of the clusters determined at random from among those clusters may be determined as the location of the new AMF 158.

(515) Then, the E2EO module 96, the configuration management module 100, and the container management module 106 constructs the new AMF 158 at the location determined in the process step of Step S705 or Step S710 (Step S711).

(516) Then, the inventory management module 98 cancels the setting of the planned attribute regarding the node 170k and the link 172j included in the topology data illustrated in FIG. 20 (Step S712), and the process illustrated in this process example is ended.

(517) [Supplement]

(518) Note that, the present invention is not limited to the embodiment described above.

(519) For example, in this embodiment, the calculation logic indicated by the calculation logic data may include a calculation expression for accumulating the performance index values of lower-level elements in the topology data and calculating the performance index value of an upper-level element. Further, in this embodiment, the performance index values of the CNFC, the NF, the NS, and the network slice may be calculated in order by tracing back the hierarchical structure indicated by the topology data. Further, the performance index value of the NSSI may be calculated based on the performance index value of the NF or the NS.

(520) Further, the performance index represented by the performance index value in this embodiment may be a throughput or a connection completion rate. Further, the performance index value may include (1) the target value or guaranteed value of the throughput, (2) the target value or guaranteed value of the connection completion rate, and (3) the target value or guaranteed value of the number of simultaneous connections.

(521) Further, the performance index value of the NS may include (1) a radio resource control (RRC) connection completion rate (KPI of RAN domain NS), (2) an attach success rate (KPI of core domain NS), and (3) an EPS radio access bearer (ERAB) established success rate (KPI of core domain NS). Further, as the performance index value (for example, connection completion rate) of an NSI (or NSSI) at a higher-level of the NS, there may be calculated the product of (1) the average value of the RRC connection completion rate, (2) the average value of the attach success rate, and (3) the average value of the ERAB established success rate. Further, the average value of the RRC connection completion rate over a plurality of gNBs belonging to the same NSI (or NSSI) may be used as the average value of the RRC connection completion rate.

Claims

1. A location determination system, comprising: at least one processor; and at least one memory device storing instructions which, when executed by the at least one processor, cause the at least one processor to perform operations comprising: storing topology data indicating a current status of a link between elements included in a communication system; storing location data indicating a current location of an element included in the communication system; and determining, in response to determination that a new element linked to an already installed element included in the communication system is to be added to the communication system, a location at which the new

element is to be constructed based on new topology data including the new element and the current location of the already installed element indicated by the location data.

2. The location determination system according to claim 1, wherein determining comprises determining a location of a cluster in which the new element is to be constructed.

3. The location determination system according to claim 2, wherein determining comprises determining a location different from a location of a cluster in which the already installed element linked to the new element in the topology data is constructed as the location of the cluster in which the new element is to be constructed.

4. The location determination system according to claim 1, wherein determining comprises determining a location of a server on which the new element is to be constructed.

5. The location determination system according to claim 4, wherein determining comprises determining a location different from a location of a server on which the already installed element linked to the new element in the topology data is constructed as the location of the server on which the new element is to be constructed.

6. The location determination system according to claim 1, wherein determining comprises determining a location of a data center in which the new element is to be constructed.

7. The location determination system according to claim 6, wherein determining comprises determining a location different from a location of a data center in which the already installed element linked to the new element in the topology data is constructed as the location of the data center in which the new element is to be constructed.

8. The location determination system according to claim 1, wherein functional units under control of a common network service are linked to each other in the topology data, wherein the location data indicates a current location of an already installed functional unit under control of an already installed network service, and wherein determining comprises determining, in response to determination that a new functional unit is to be added under the control of the already installed network service included in the communication system, a location at which the new functional unit is to be constructed based on the current location of the already installed functional unit indicated as the functional unit under the control of the already installed network service by the topology data, the current location being indicated by the location data.

9. The location determination system according to claim 1, wherein functional units belonging to a common network slice are linked to each other in the topology data, wherein the location data indicates a current location of an already installed functional unit belonging to an already installed network slice, and wherein determining comprises determining, in response to determination that a new functional unit belonging to the already installed network slice included in the communication system is to be added to the communication system, a location at which the new functional unit is to be constructed based on the current location of the already installed functional unit indicated as the functional unit belonging to the already installed network slice by the topology data, the current location being indicated by the location data.

10. The location determination system according to claim 1, wherein determining comprises determining a location satisfying a given performance requirement relating to communication between the already installed element and the new element as the location at which the new element is to be constructed.

11. The location determination system according to claim 1, wherein the operations further comprise constructing the new element at the determined location.

12. A location determination method, comprising: storing topology data indicating a current status of a link between elements included in a communication system; storing location data indicating a current location of an element included in the communication system; and determining, in response to determination that a new element linked to an already installed element included in the communication system is to be added to the communication system, a location at which the new

element is to be constructed based on new topology data including the new element and the current location of the already installed element indicated by the location data.
